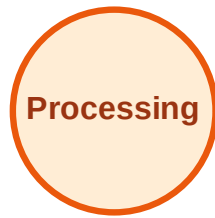


BFS Traversal

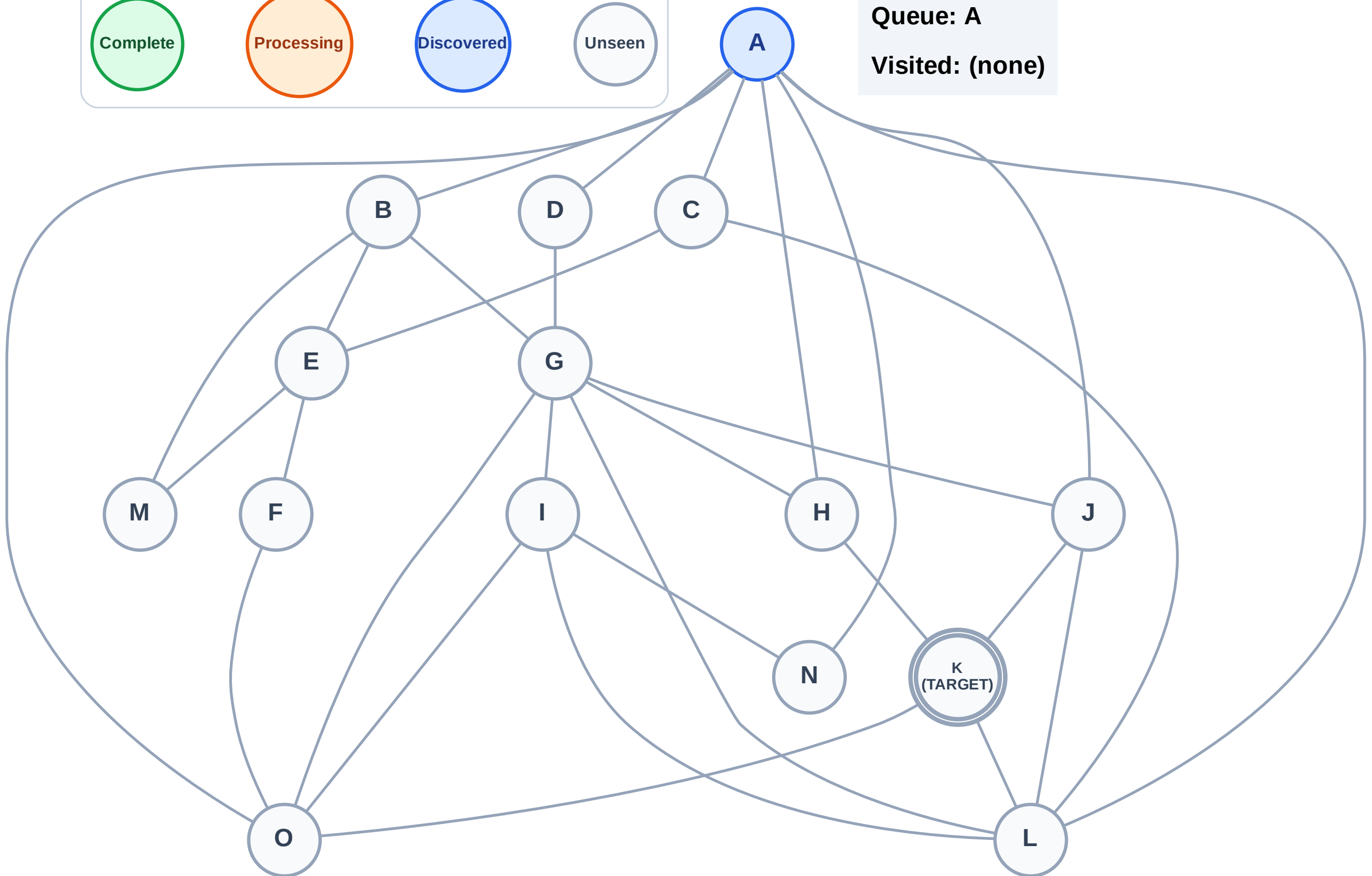
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



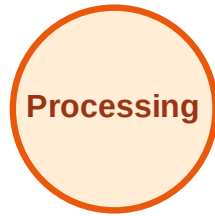
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



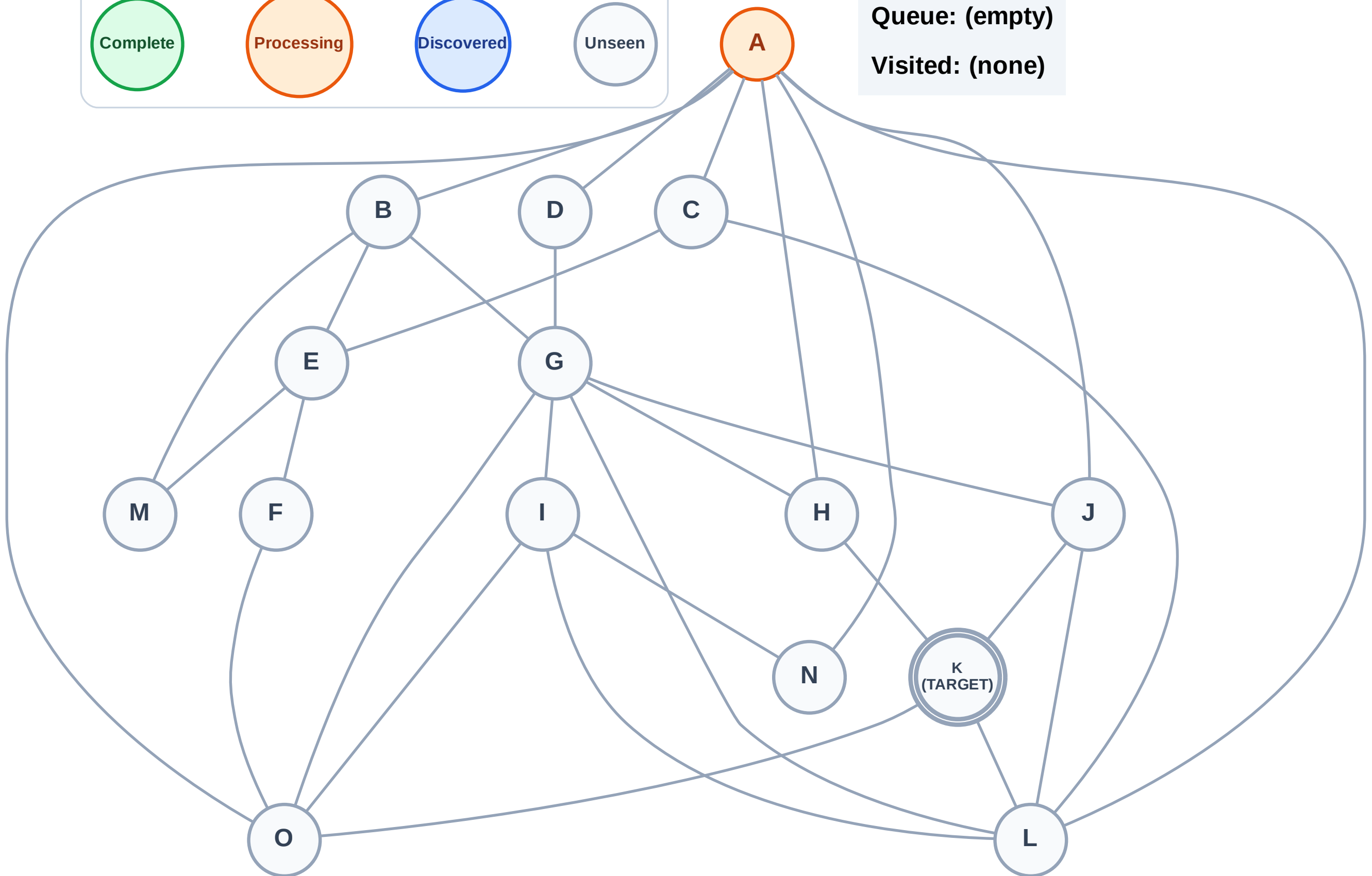
Discovered



Unseen

Queue: (empty)

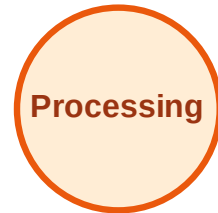
Visited: (none)



BFS Traversal

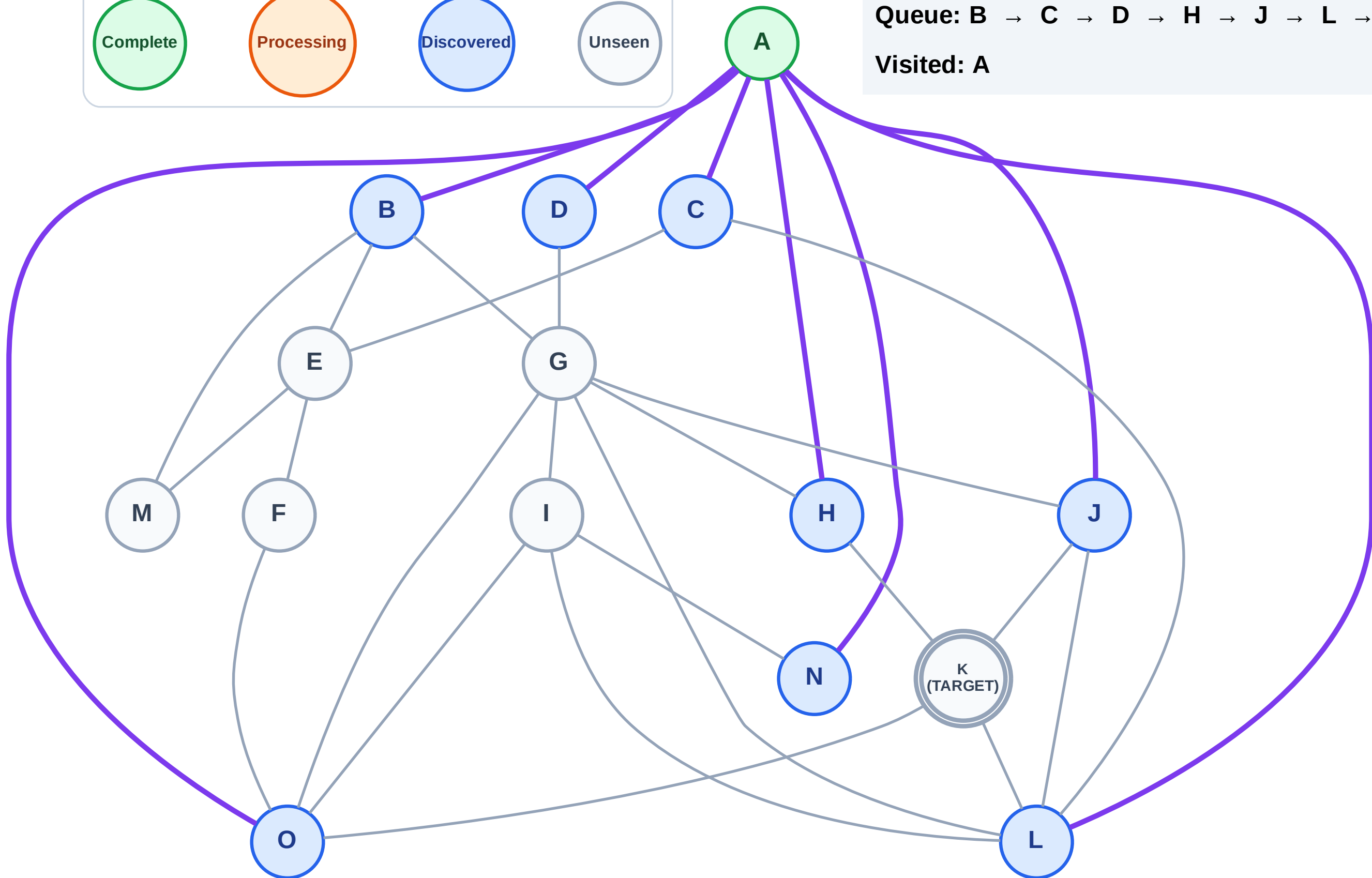
Step 3: Discover B, C, D, H, J, L, N, O from A

Legend



Queue: B → C → D → H → J → L → N → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

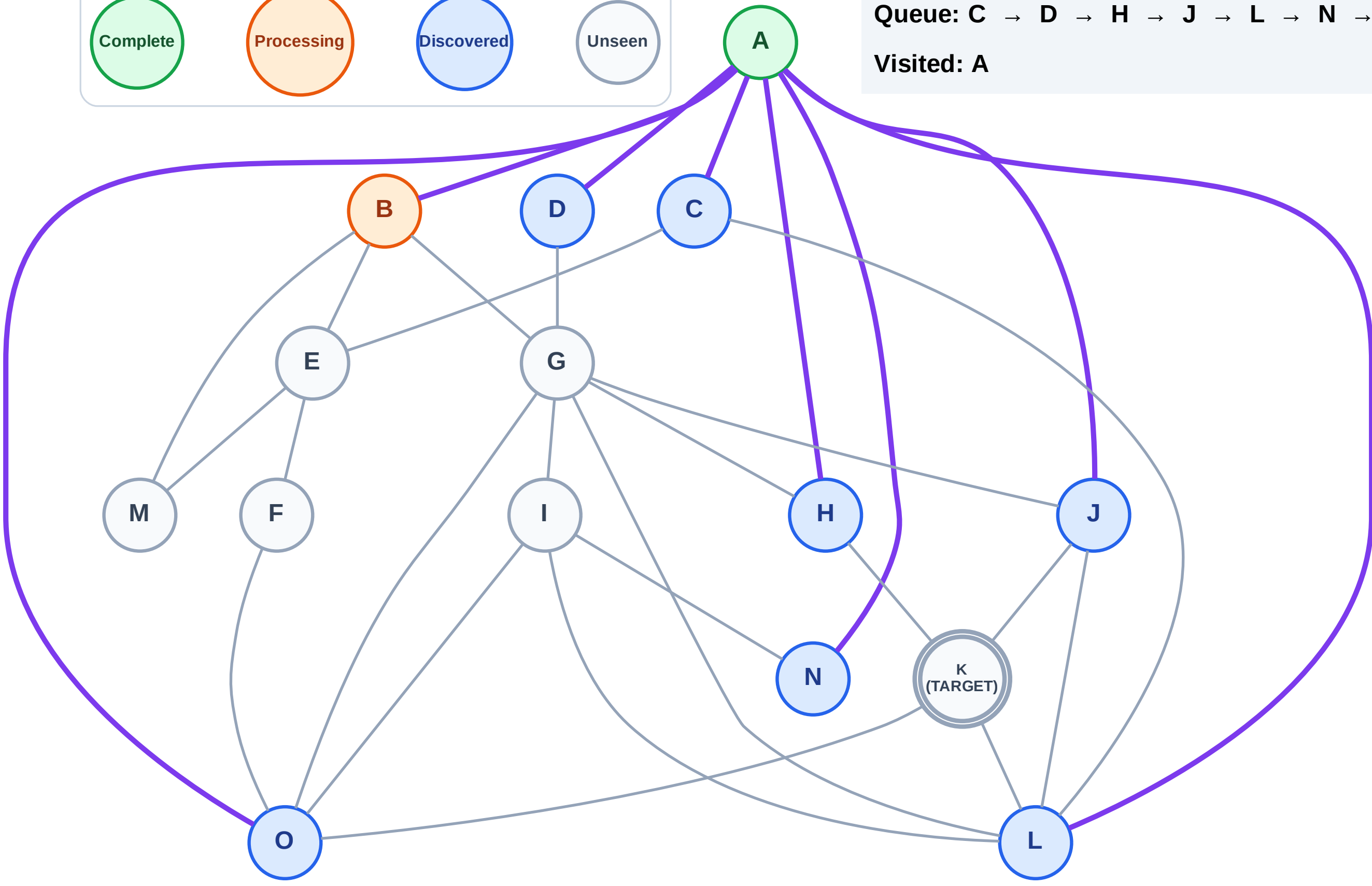
Processing

Discovered

Unseen

Queue: C → D → H → J → L → N → O

Visited: A



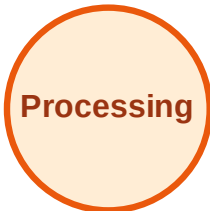
BFS Traversal

Step 5: Discover E, G, M from B

Legend



Complete



Processing



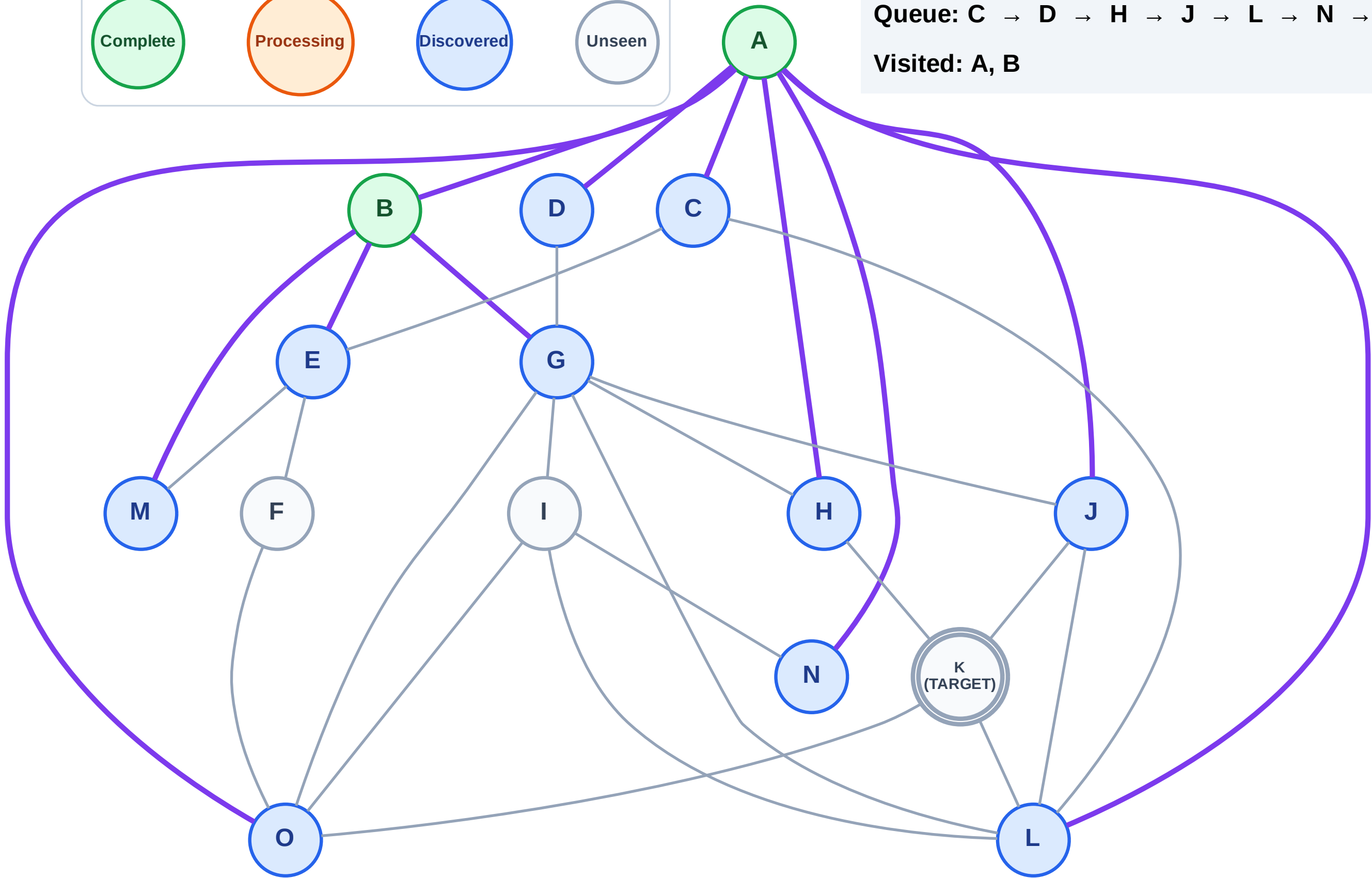
Discovered



Unseen

Queue: C → D → H → J → L → N → O → E → G → M

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

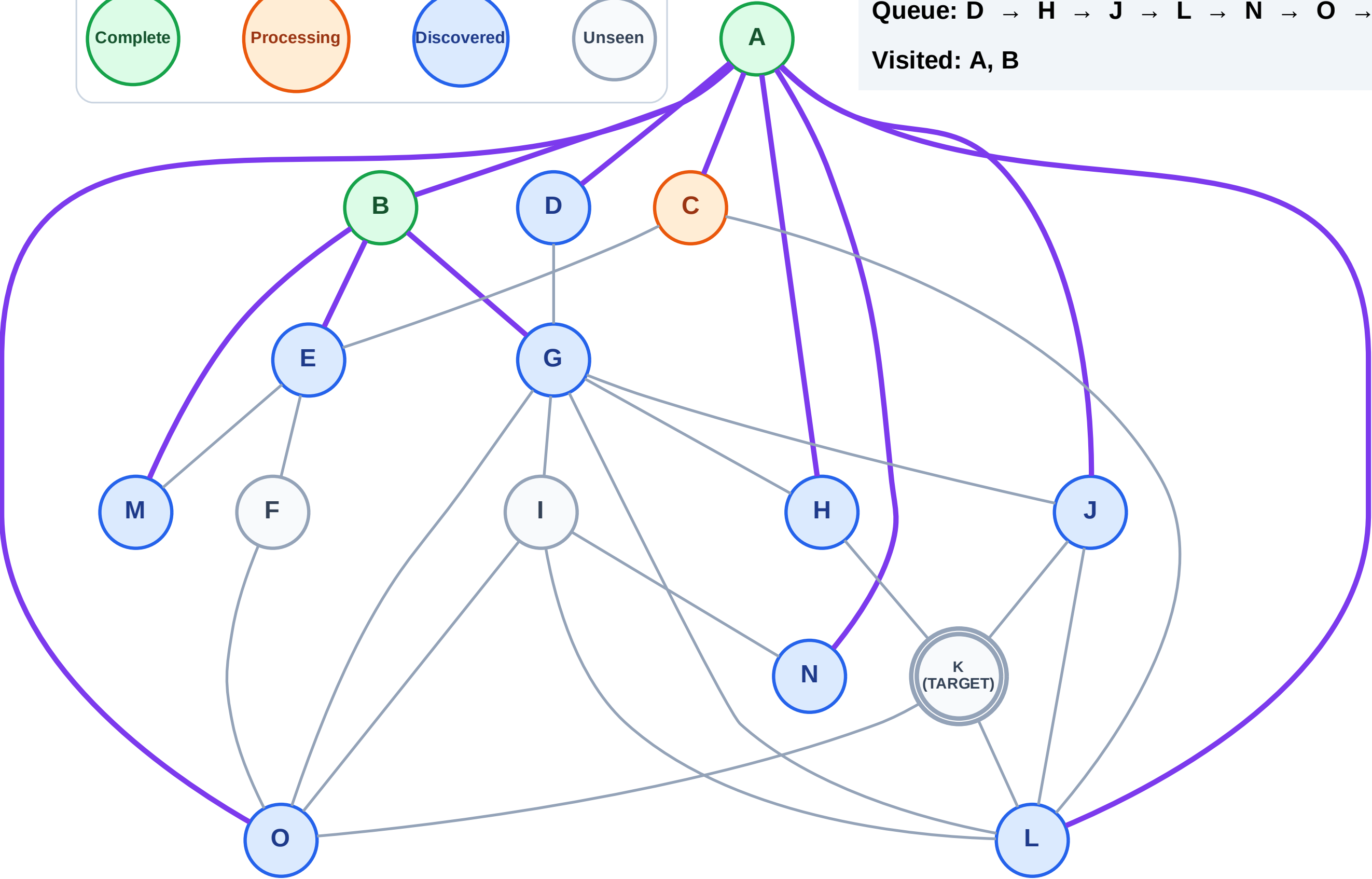
Processing

Discovered

Unseen

Queue: D → H → J → L → N → O → E → G → M

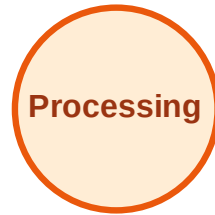
Visited: A, B



BFS Traversal

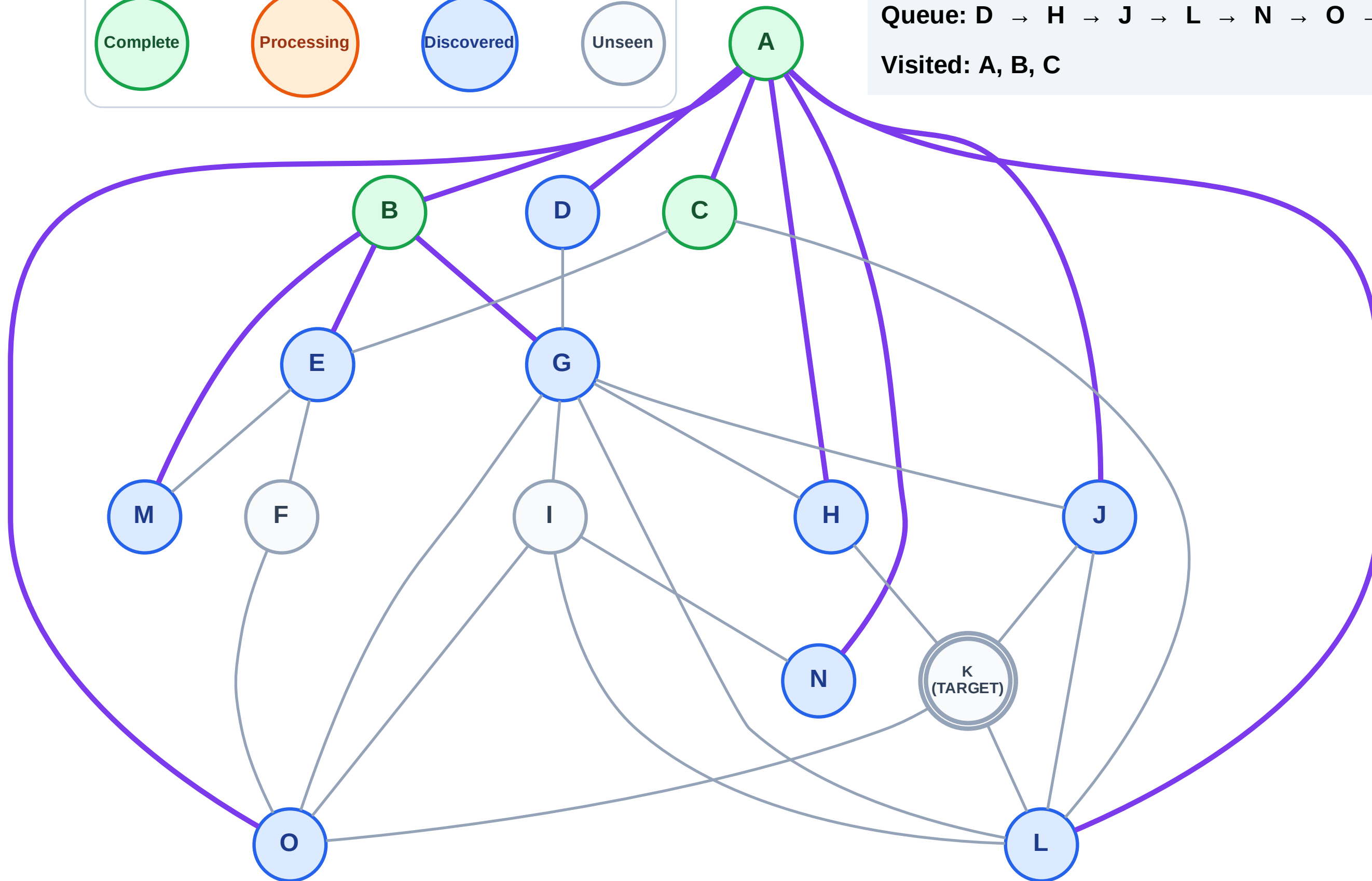
Step 7: No new neighbors from C

Legend



Queue: D → H → J → L → N → O → E → G → M

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

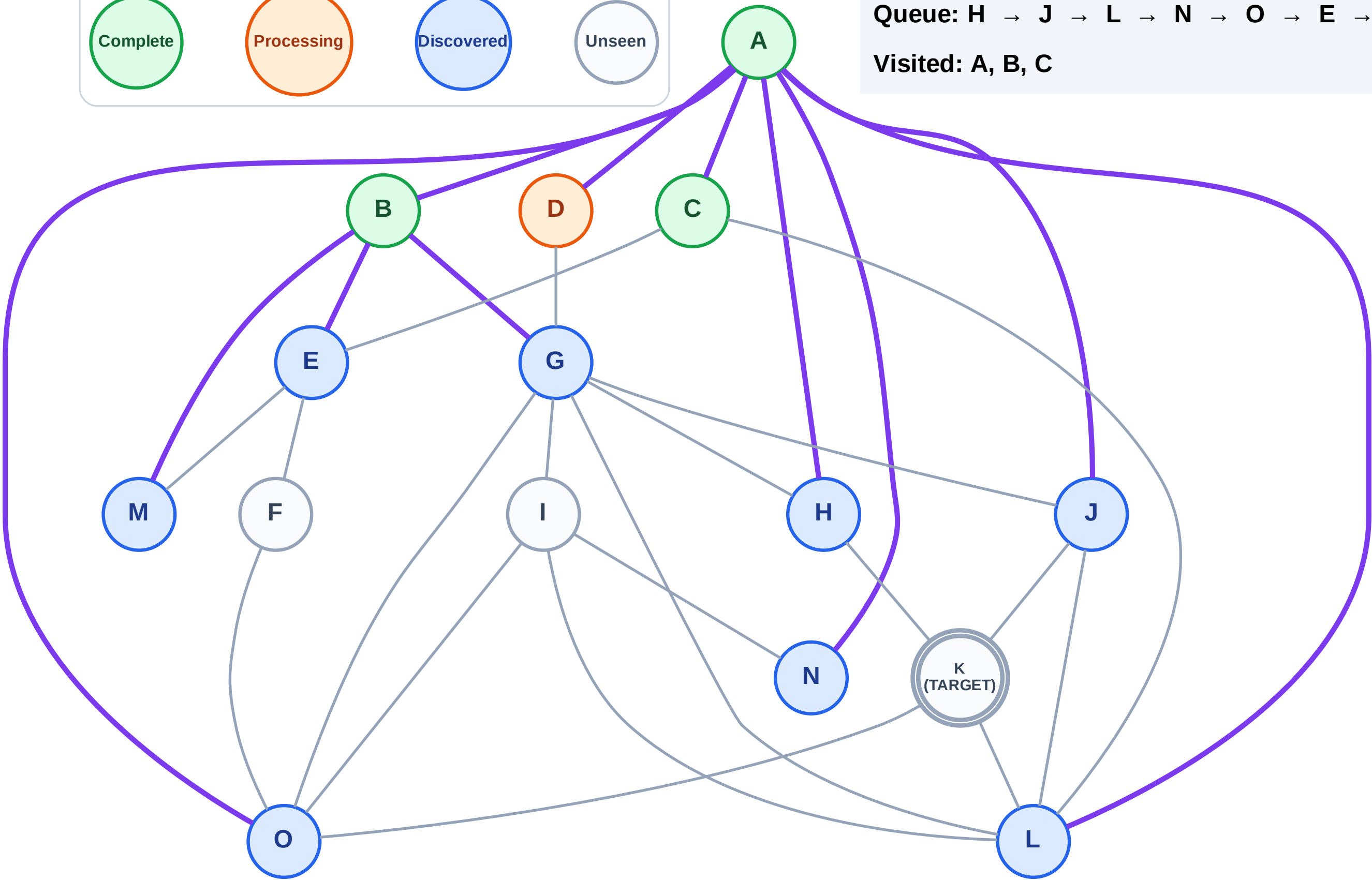
Processing

Discovered

Unseen

Queue: H → J → L → N → O → E → G → M

Visited: A, B, C



BFS Traversal

Step 9: No new neighbors from D

Legend

Complete

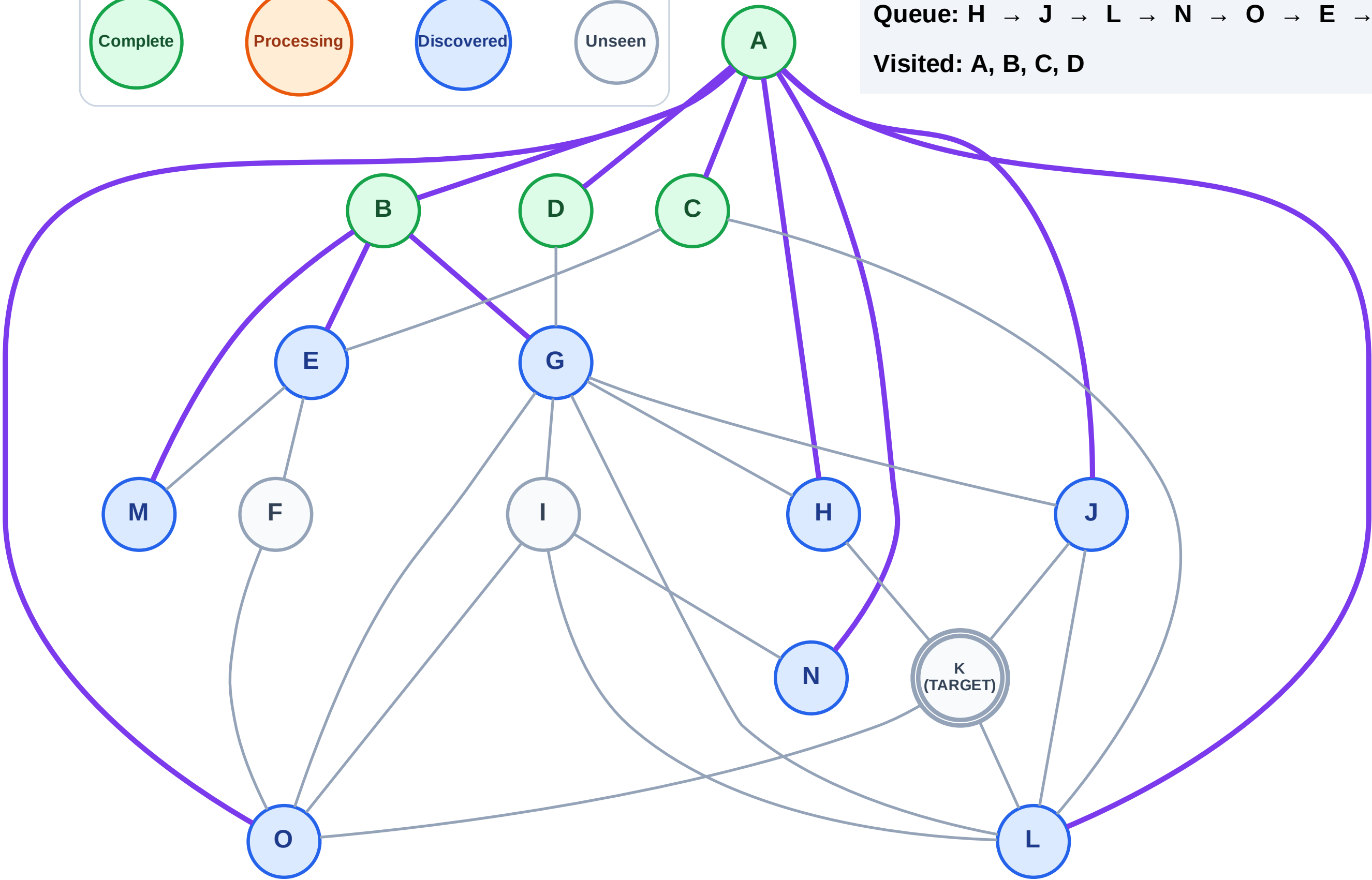
Processing

Discovered

Unseen

Queue: H → J → L → N → O → E → G → M

Visited: A, B, C, D



BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



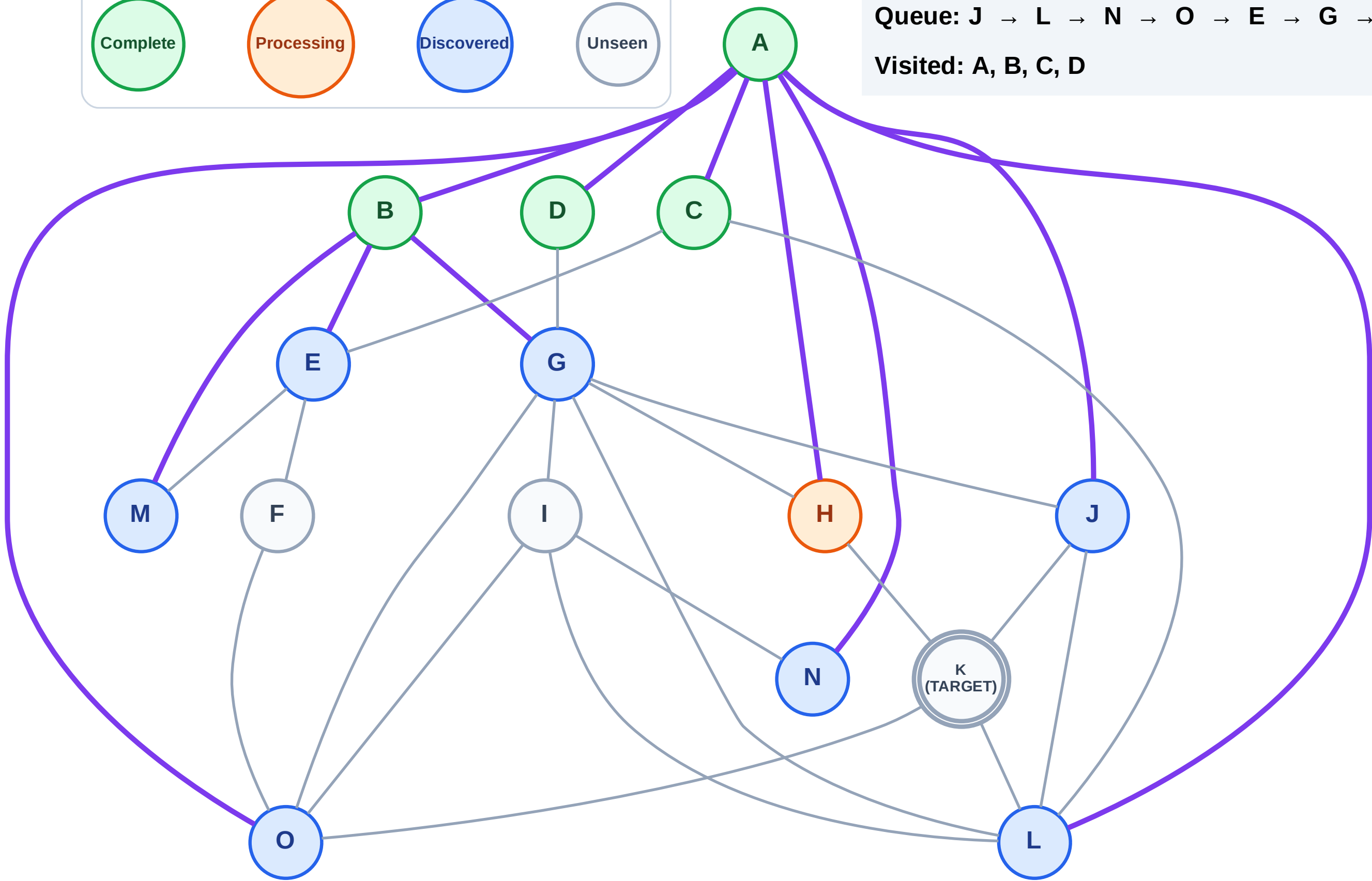
Discovered



Unseen

Queue: J → L → N → O → E → G → M

Visited: A, B, C, D



BFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → H → K

Legend

Complete

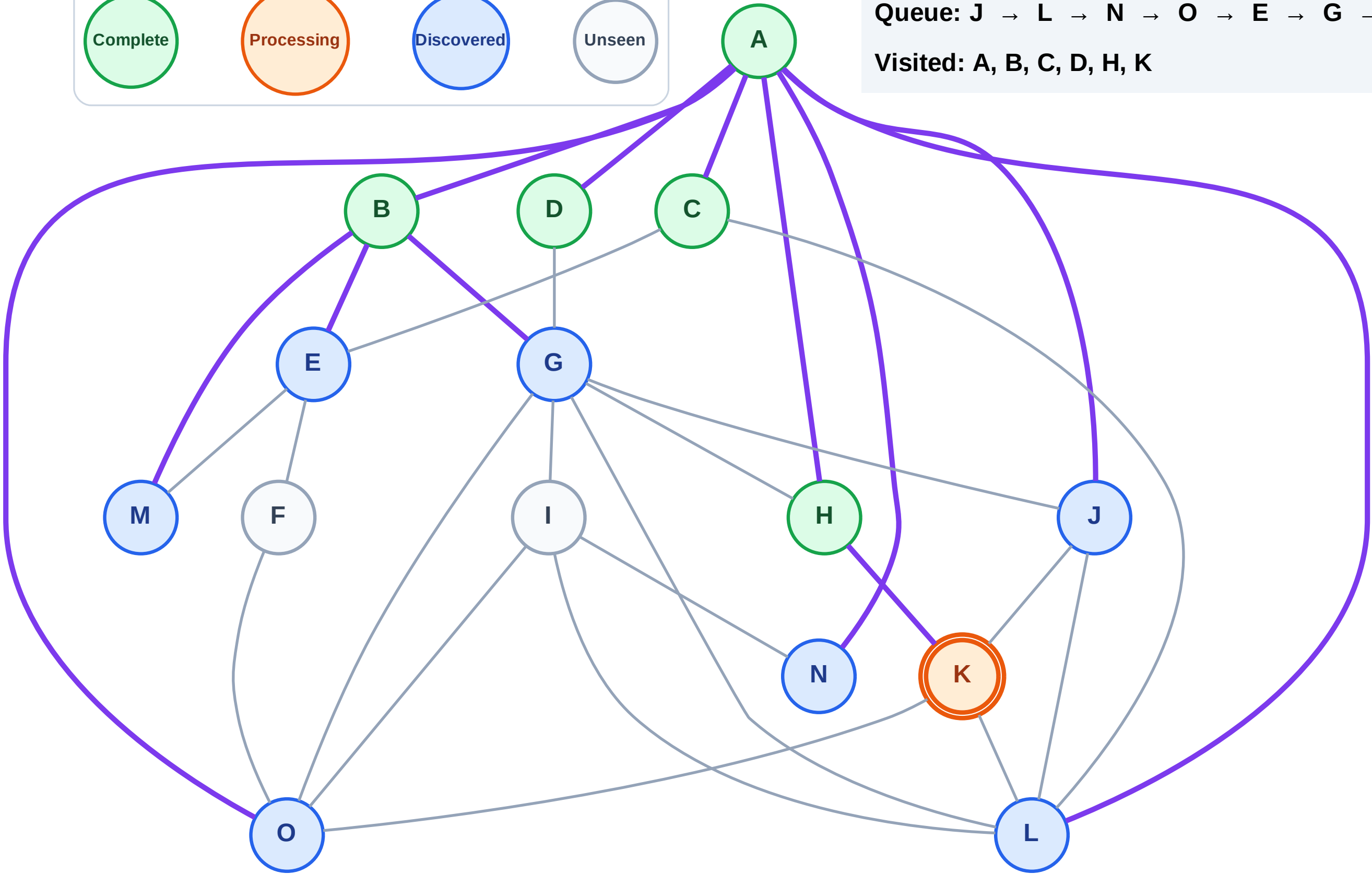
Processing

Discovered

Unseen

Queue: J → L → N → O → E → G → M

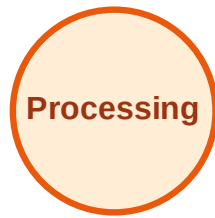
Visited: A, B, C, D, H, K



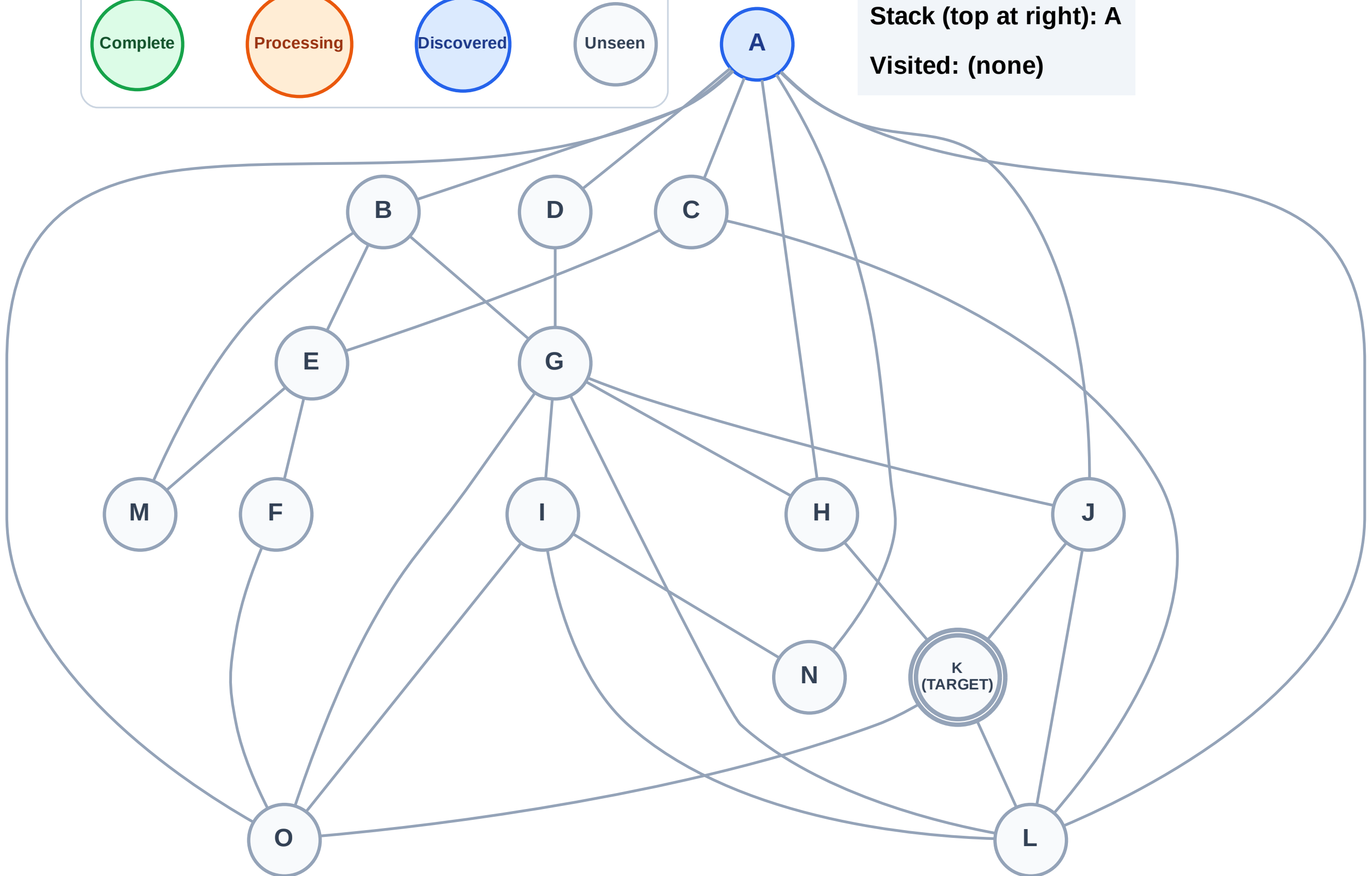
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



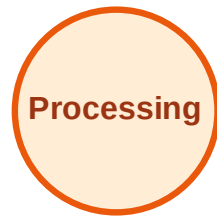
Stack (top at right): A
Visited: (none)



DFS Traversal

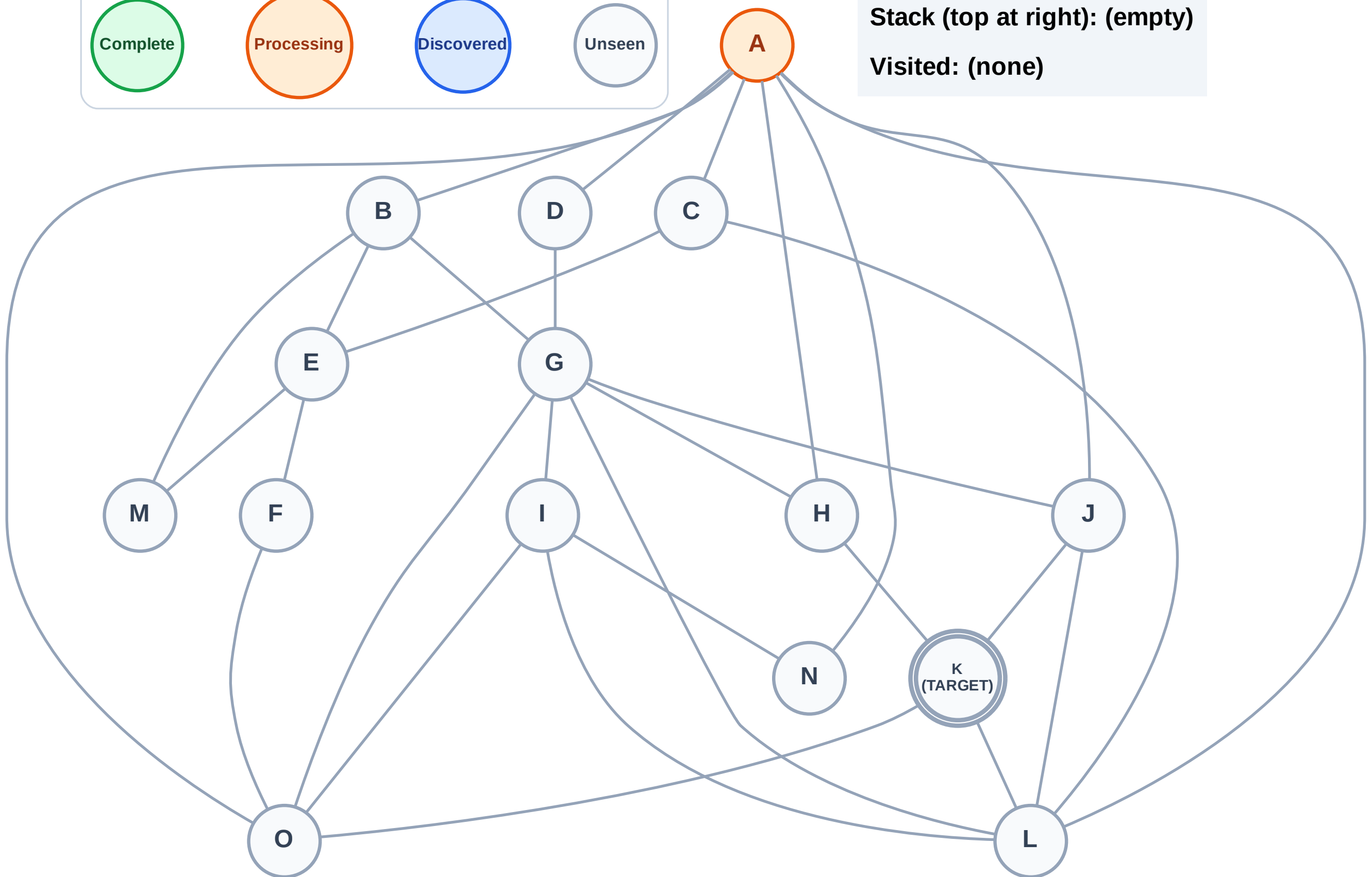
Step 2: Process node A

Legend



Stack (top at right): (empty)

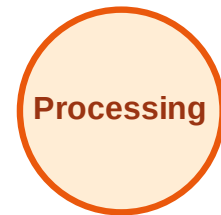
Visited: (none)



DFS Traversal

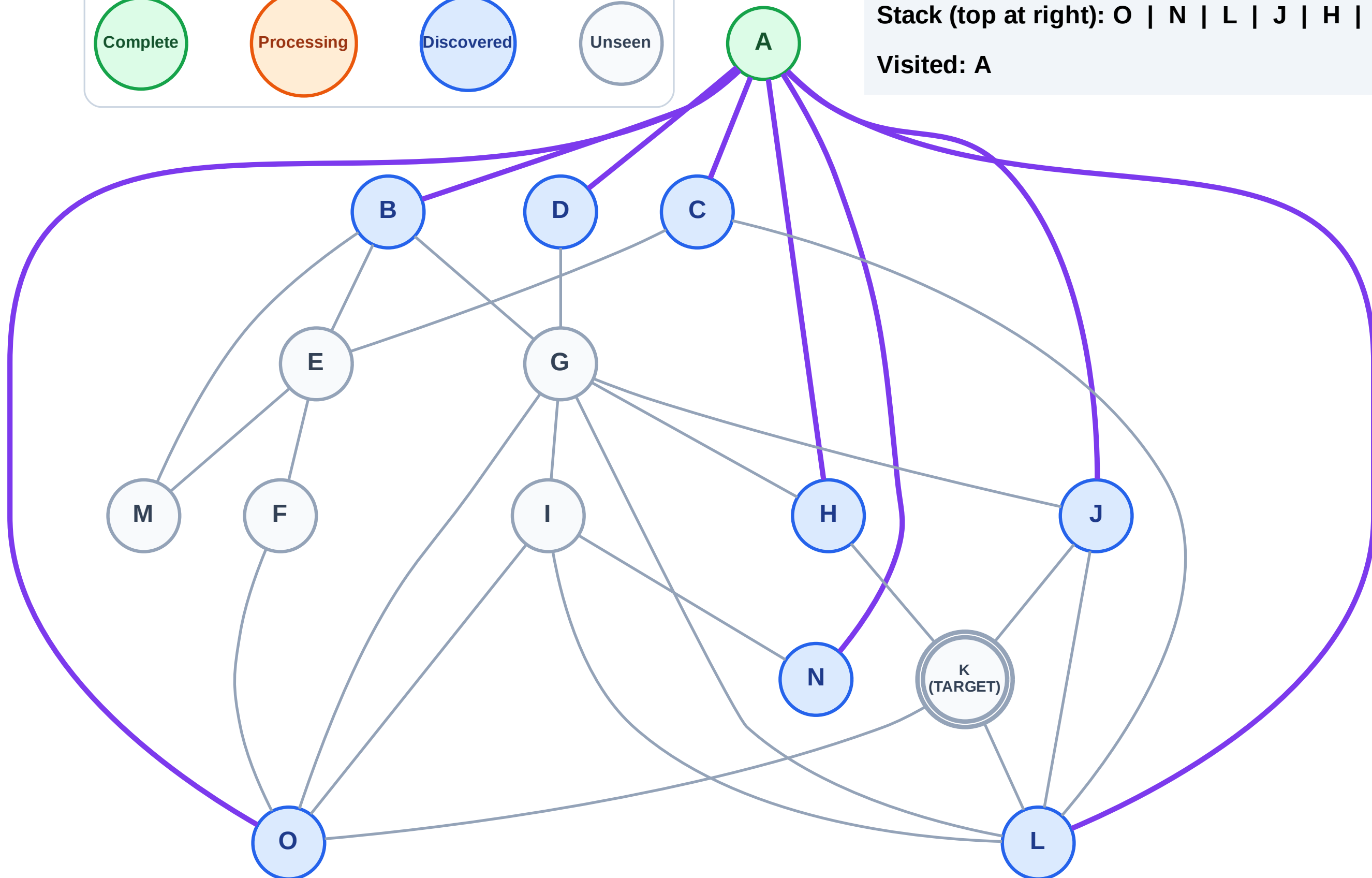
Step 3: Discover O, N, L, J, H, D, C, B from A

Legend



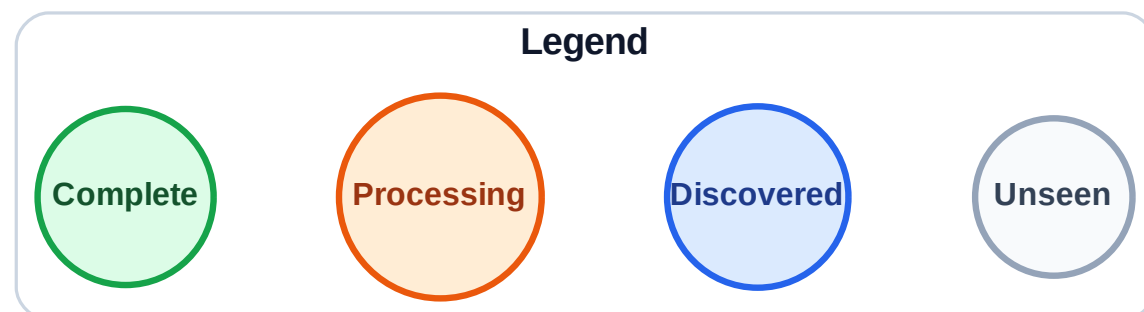
Stack (top at right): O | N | L | J | H | D | C | B

Visited: A

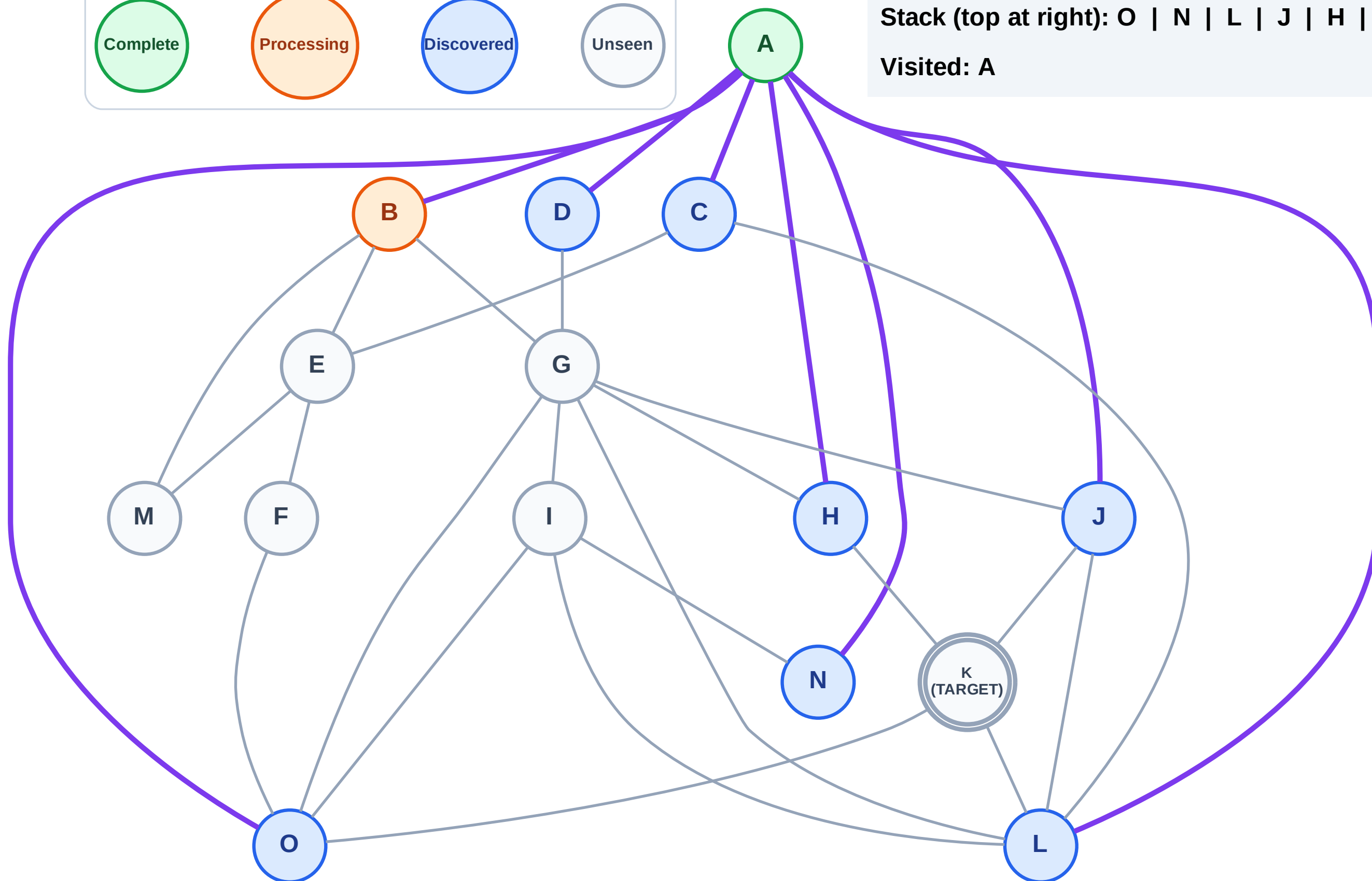


DFS Traversal

Step 4: Process node B



Stack (top at right): O | N | L | J | H | D | C
Visited: A



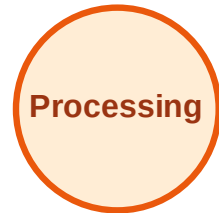
DFS Traversal

Step 5: Discover M, G, E from B

Legend



Complete



Processing



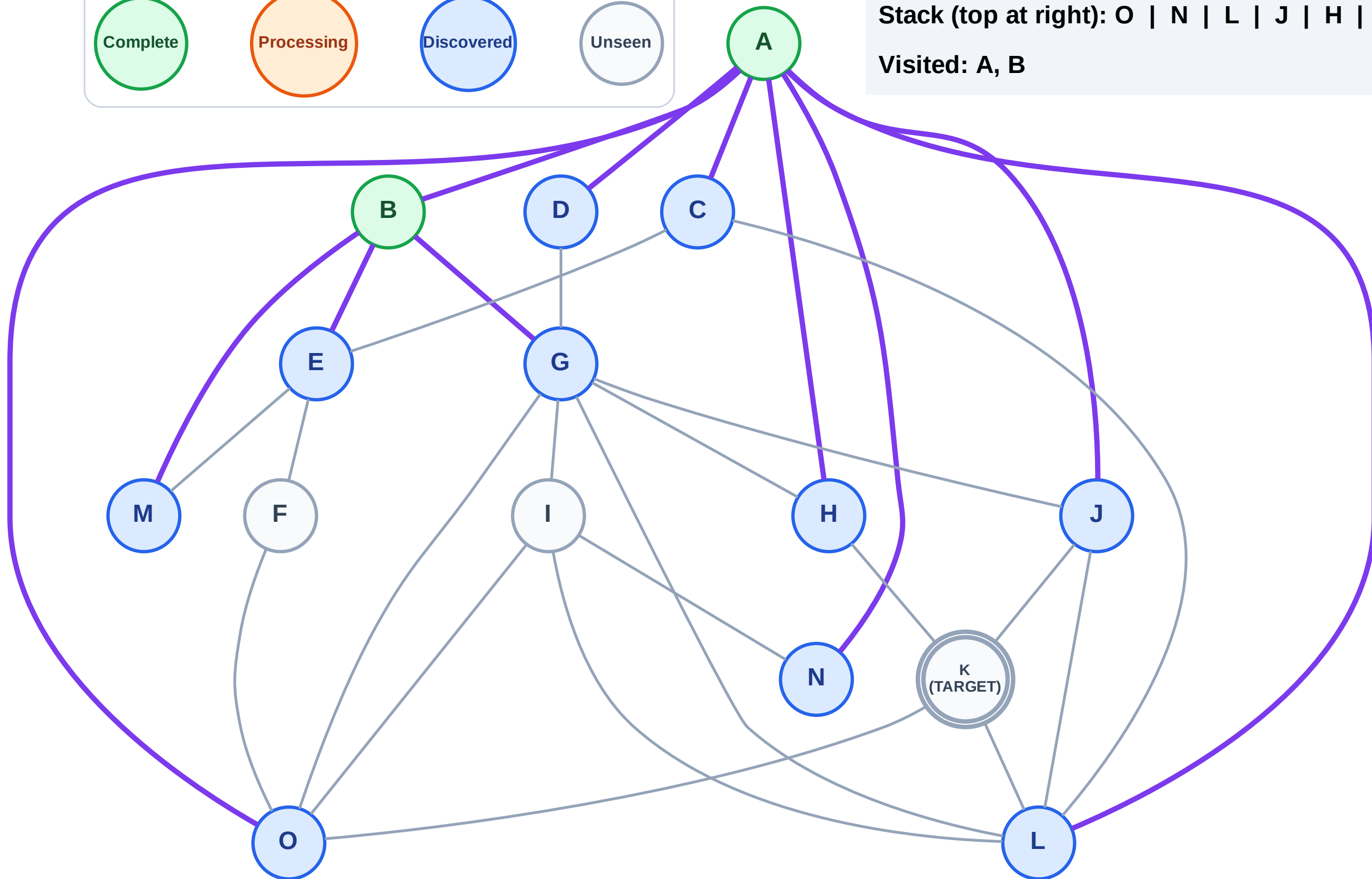
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D | C | M | G | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

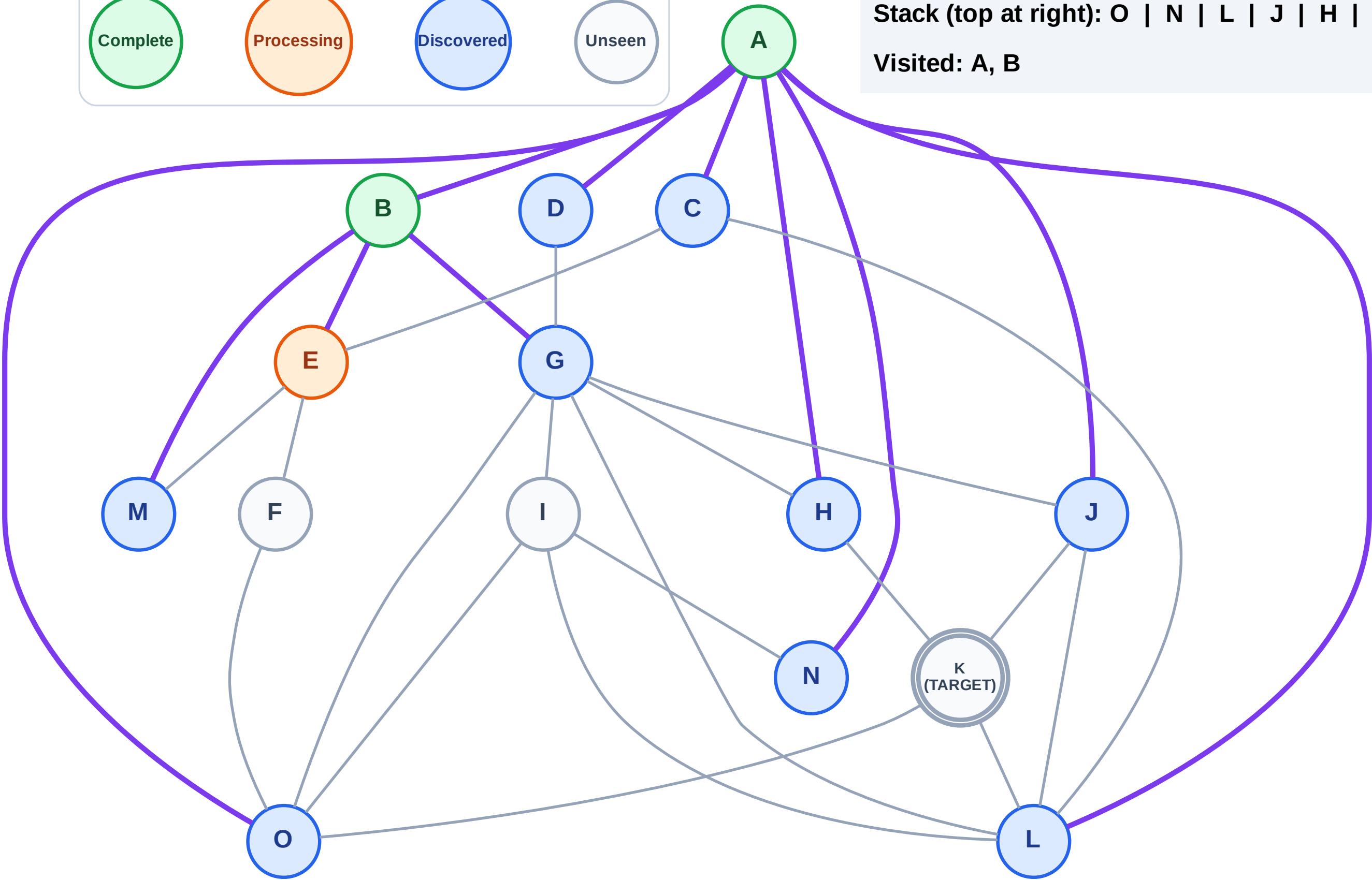
Processing

Discovered

Unseen

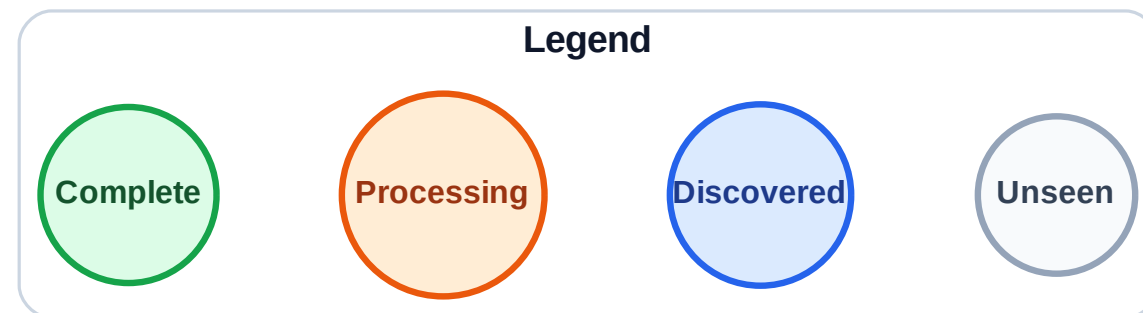
Stack (top at right): O | N | L | J | H | D | C | M | G

Visited: A, B



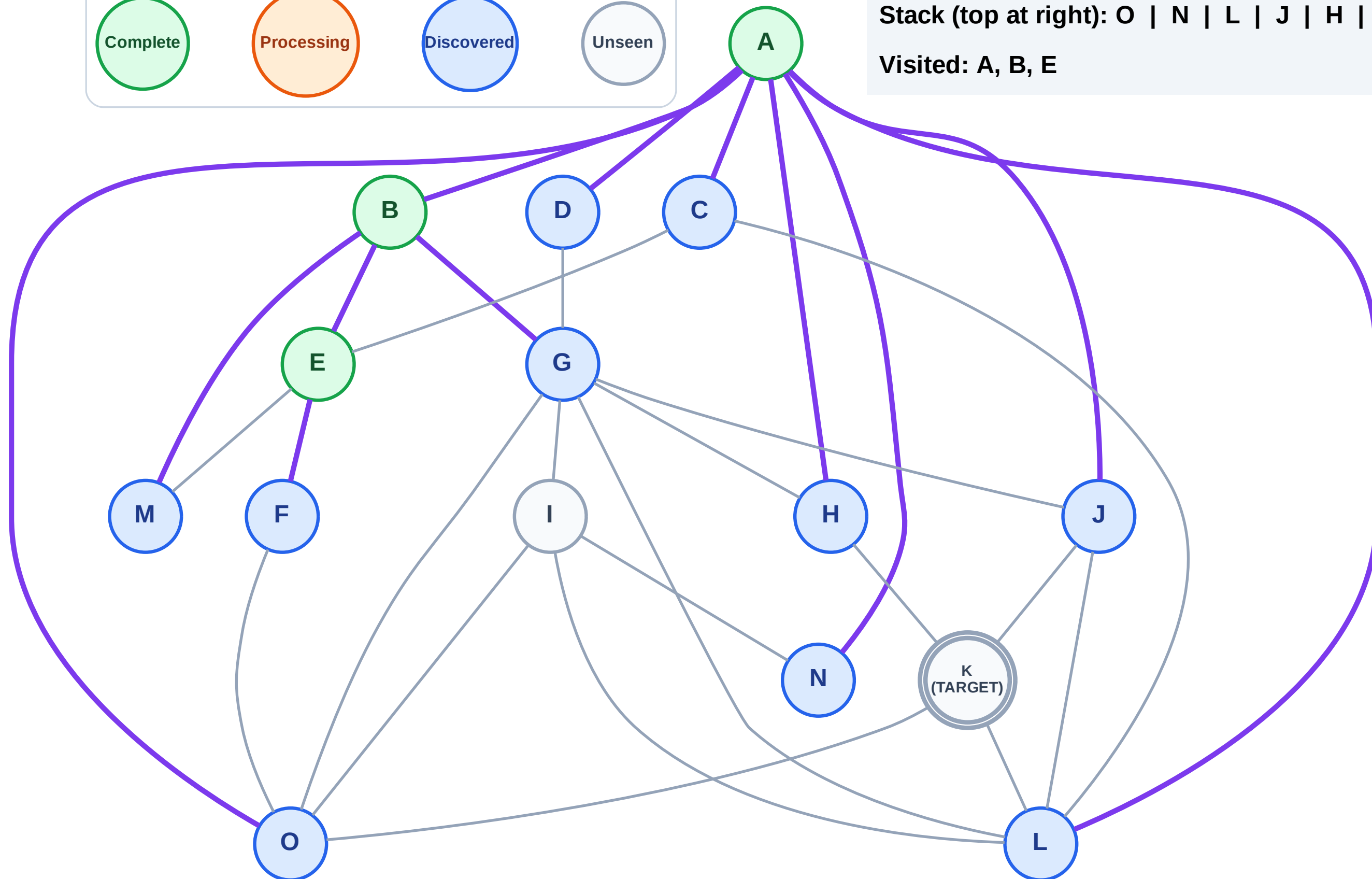
DFS Traversal

Step 7: Discover F from E



Stack (top at right): O | N | L | J | H | D | C | M | G | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

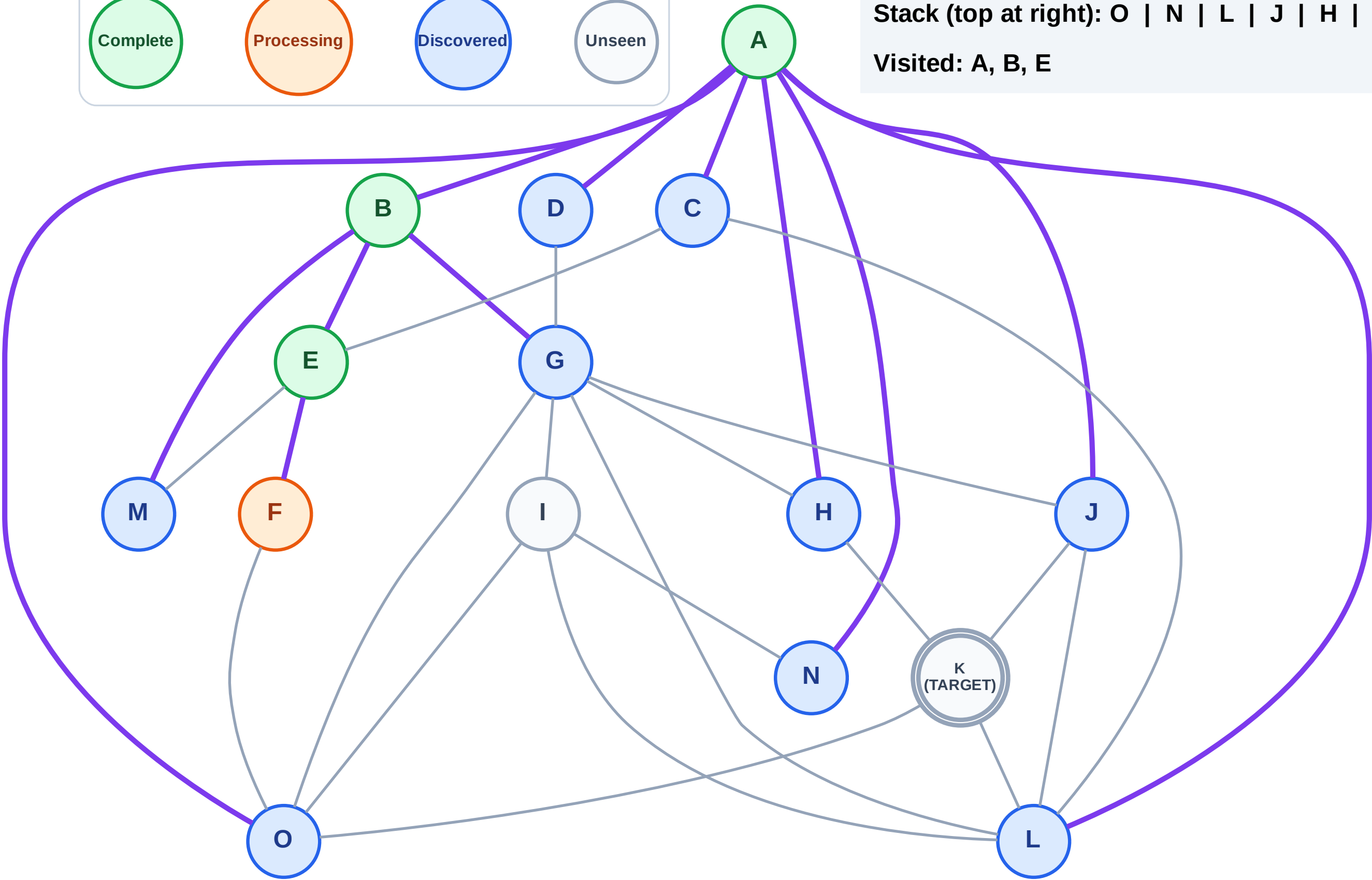
Processing

Discovered

Unseen

Stack (top at right): O | N | L | J | H | D | C | M | G

Visited: A, B, E



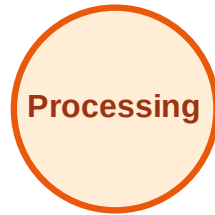
DFS Traversal

Step 9: No new neighbors from F

Legend



Complete



Processing



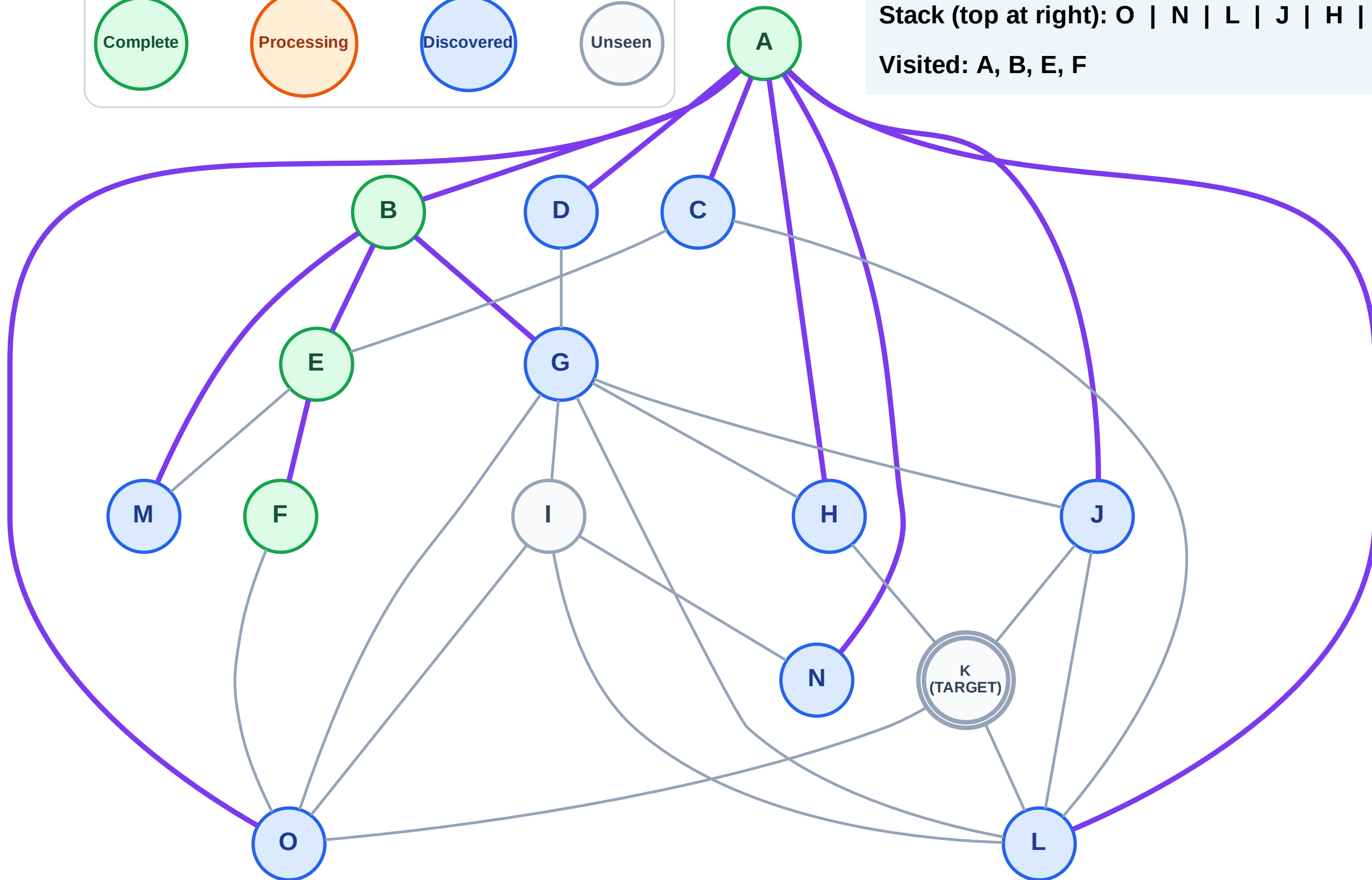
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D | C | M | G

Visited: A, B, E, F



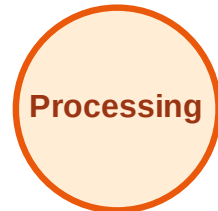
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



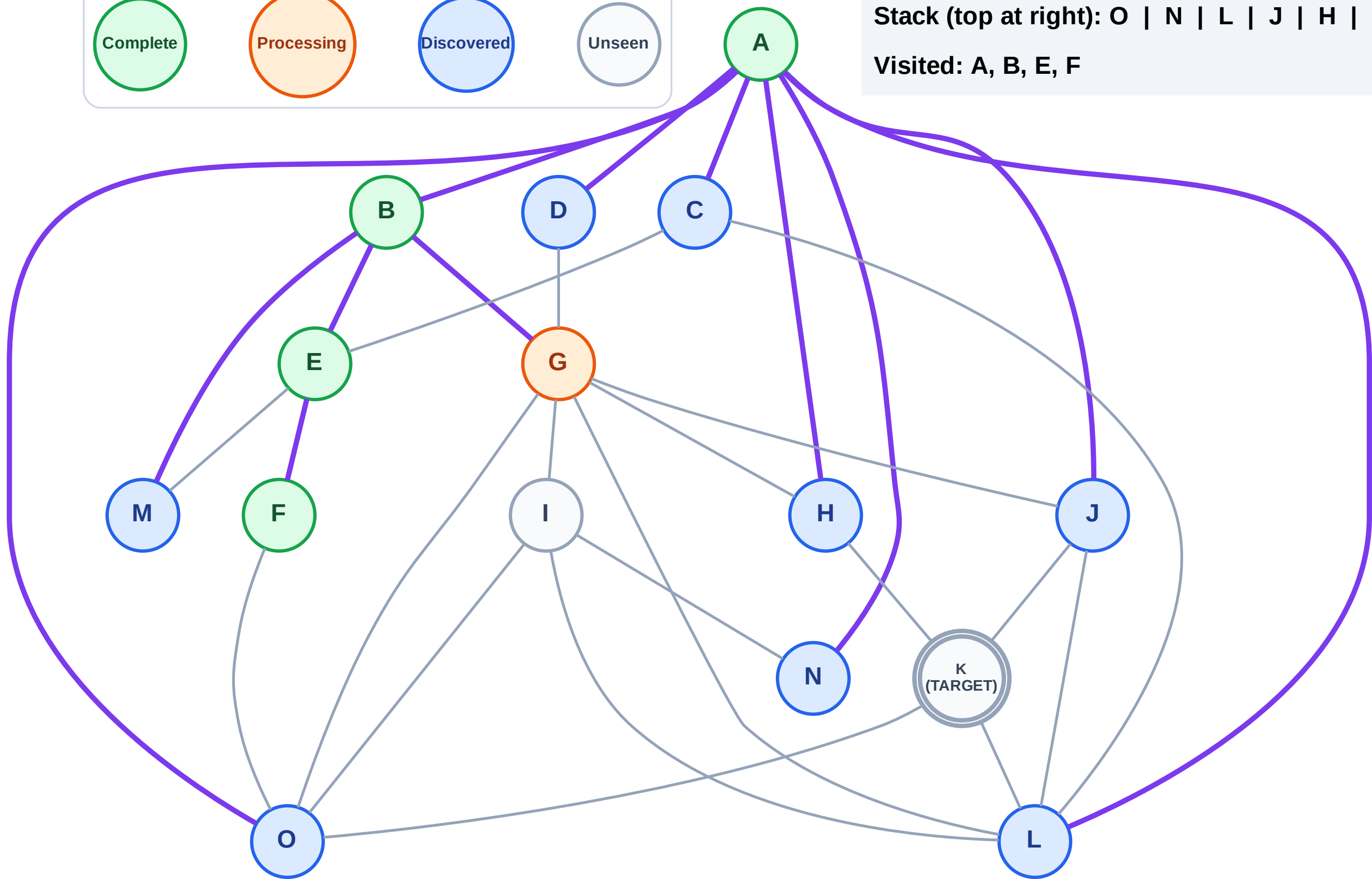
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D | C | M

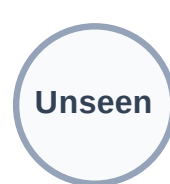
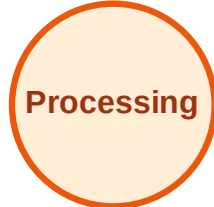
Visited: A, B, E, F



DFS Traversal

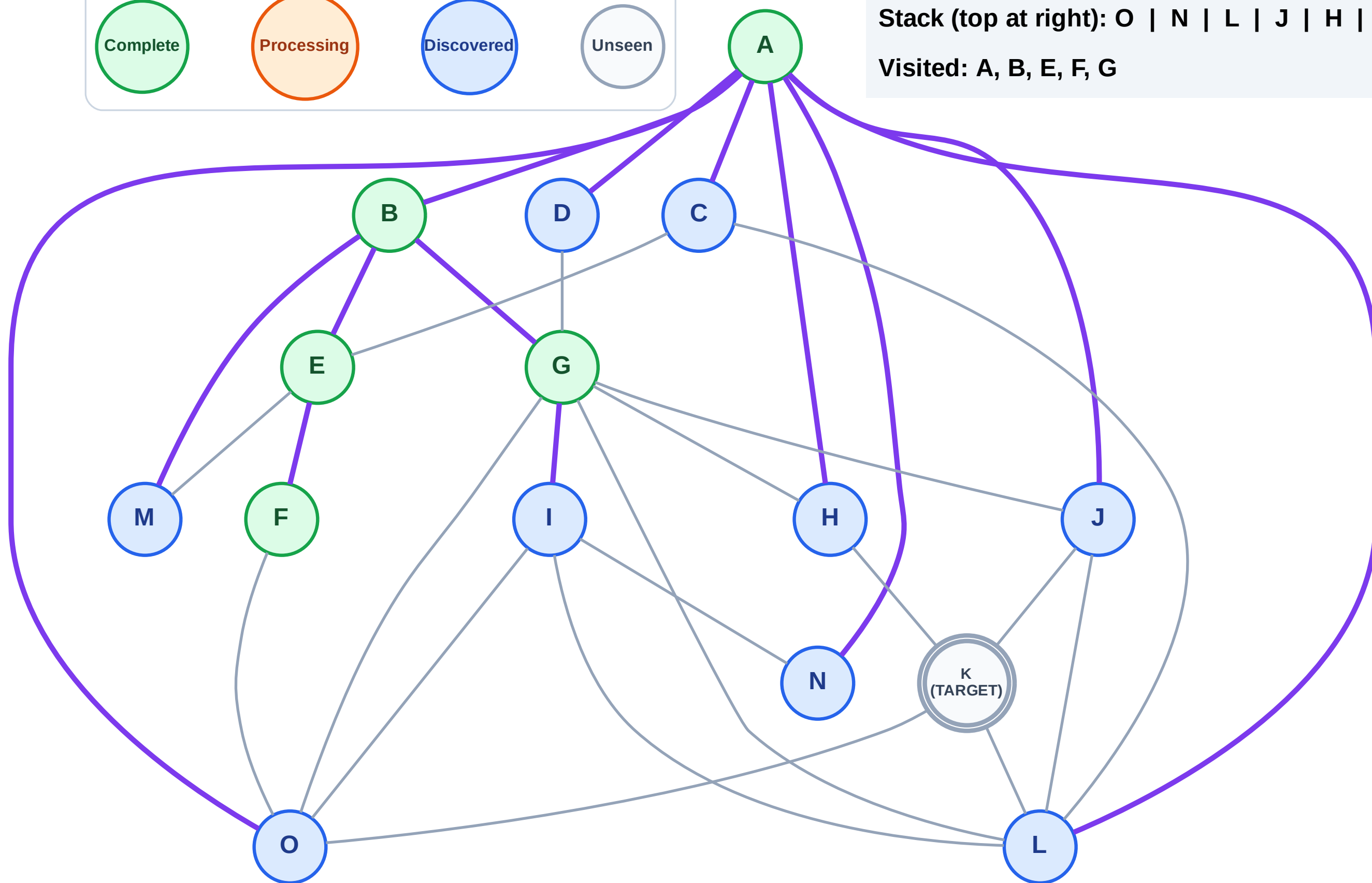
Step 11: Discover I from G

Legend

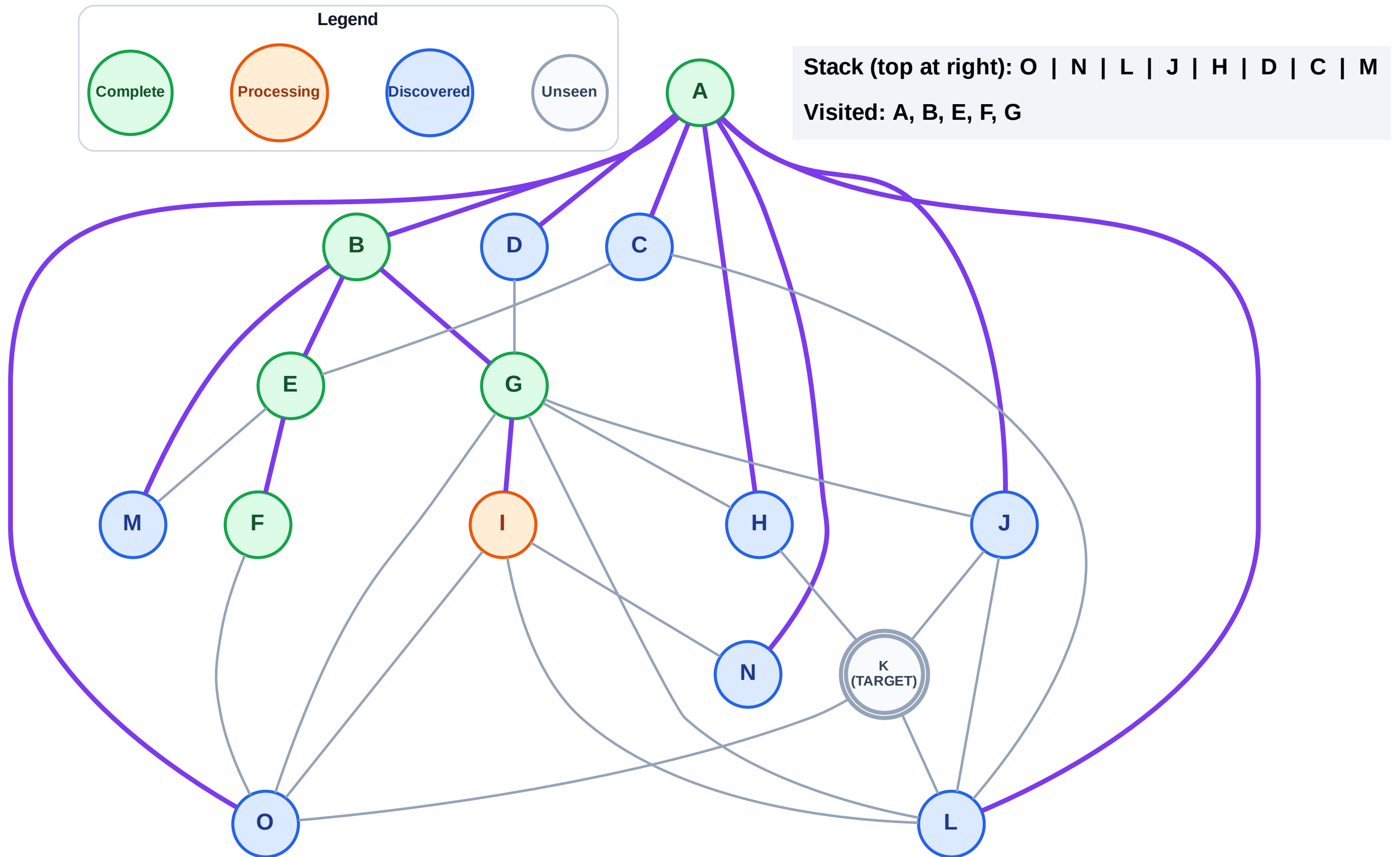


Stack (top at right): O | N | L | J | H | D | C | M | I

Visited: A, B, E, F, G



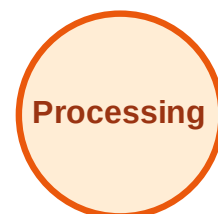
Step 12: Process node I



DFS Traversal

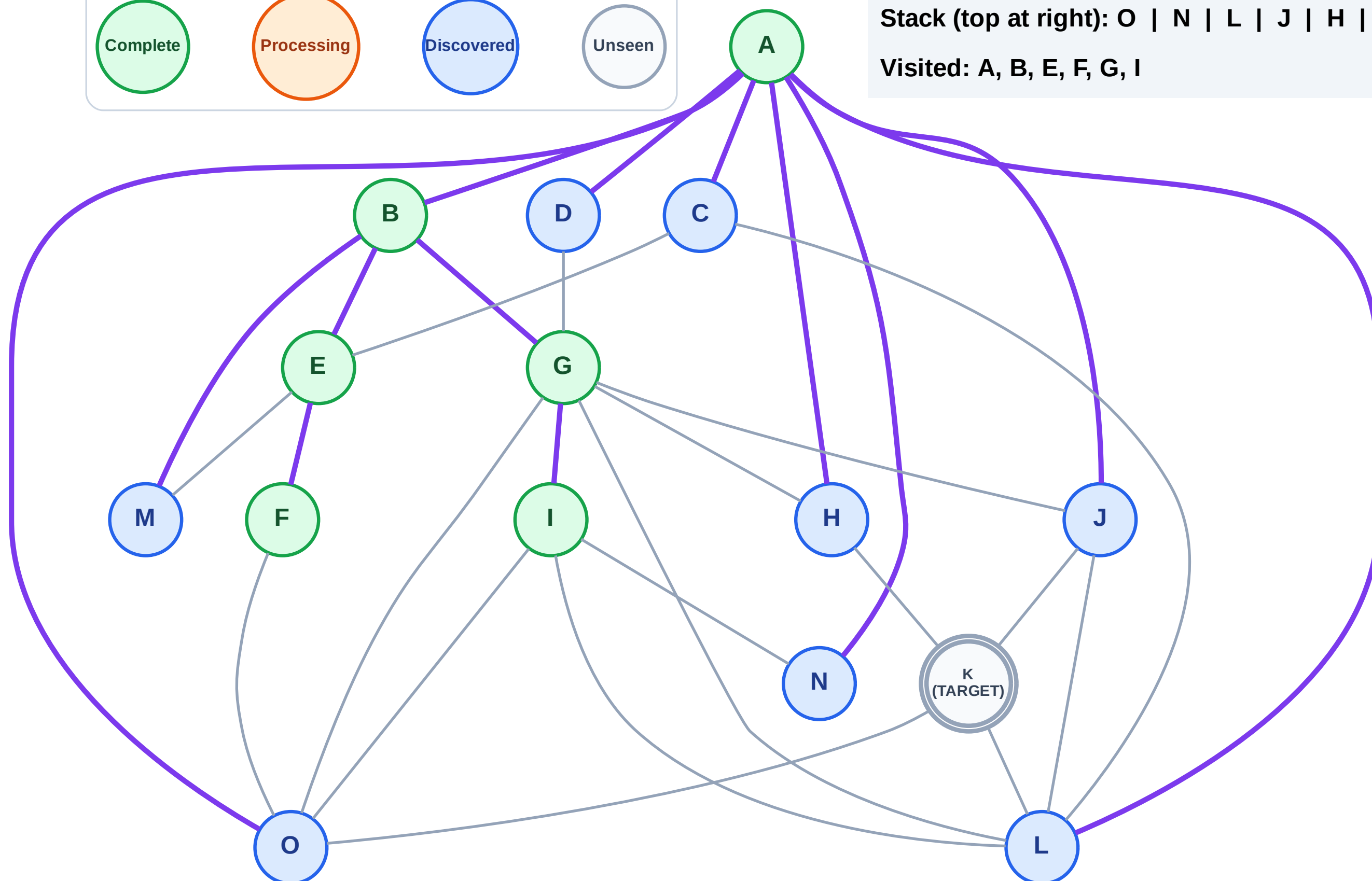
Step 13: No new neighbors from I

Legend



Stack (top at right): O | N | L | J | H | D | C | M

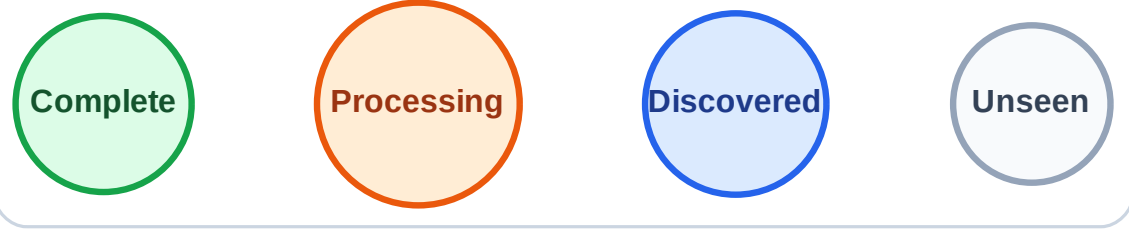
Visited: A, B, E, F, G, I



DFS Traversal

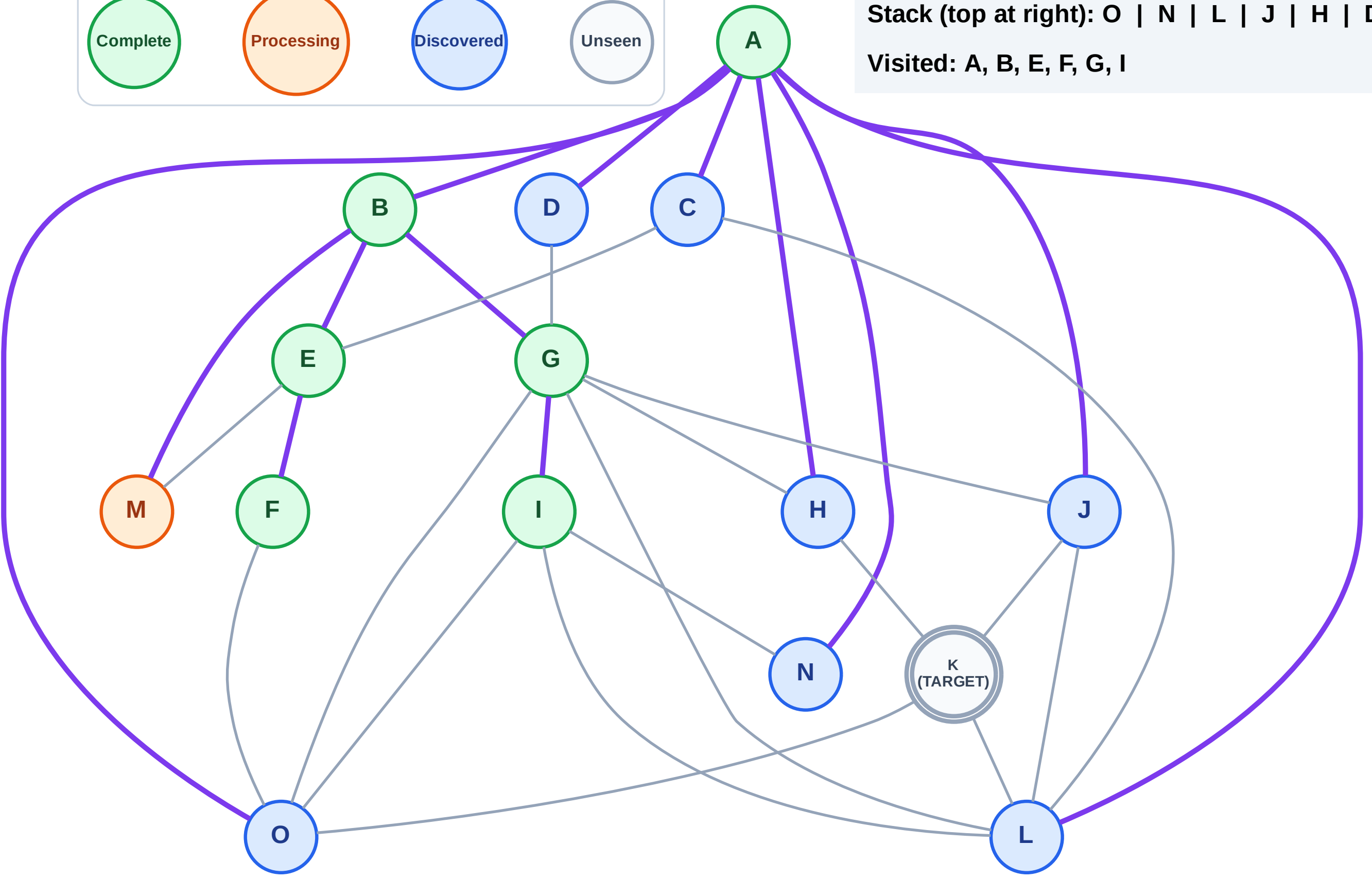
Step 14: Process node M

Legend



Stack (top at right): O | N | L | J | H | D | C

Visited: A, B, E, F, G, I



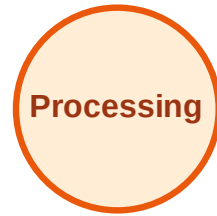
DFS Traversal

Step 15: No new neighbors from M

Legend



Complete



Processing



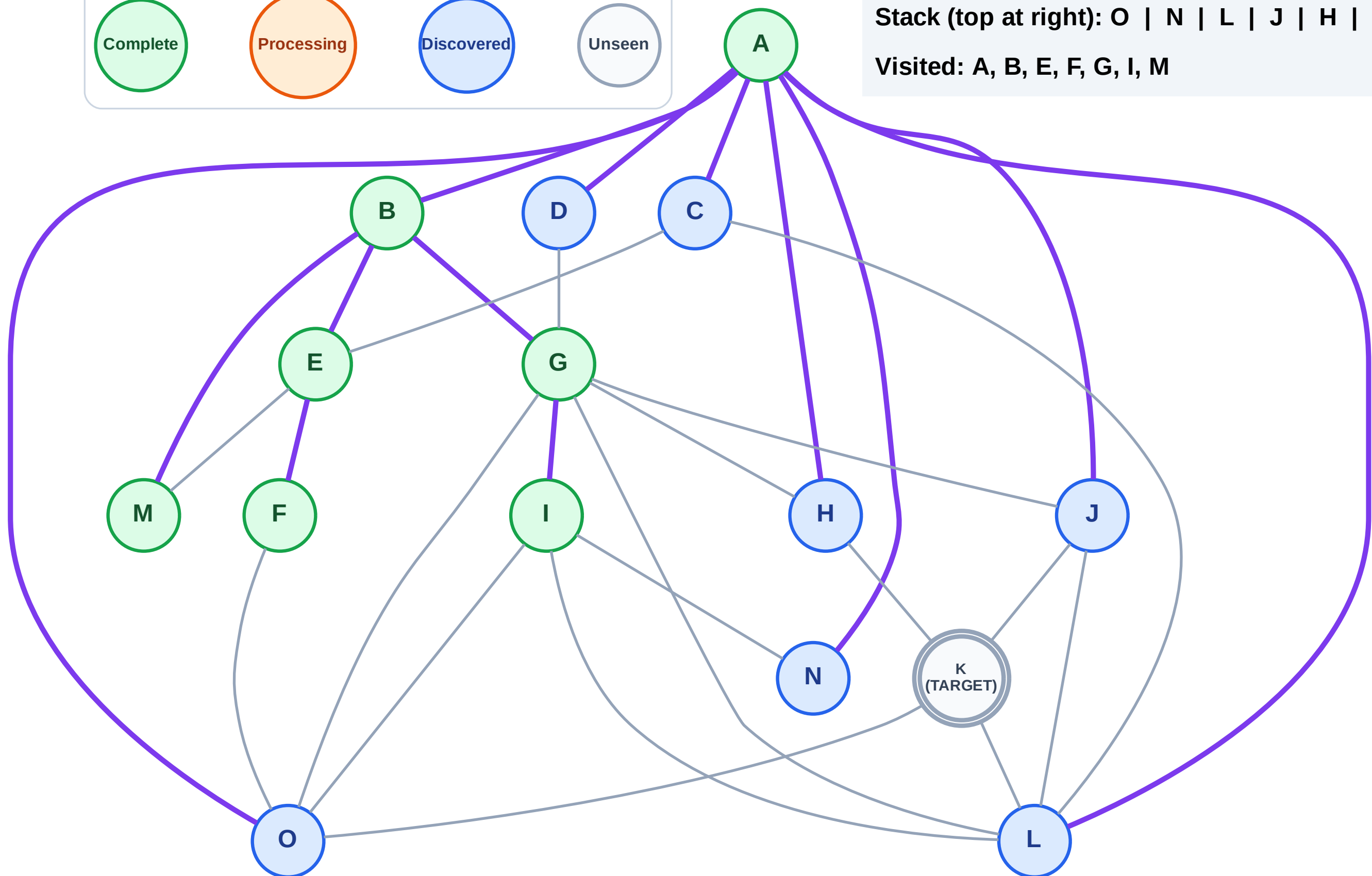
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D | C

Visited: A, B, E, F, G, I, M



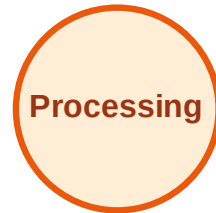
DFS Traversal

Step 16: Process node C

Legend



Complete



Processing



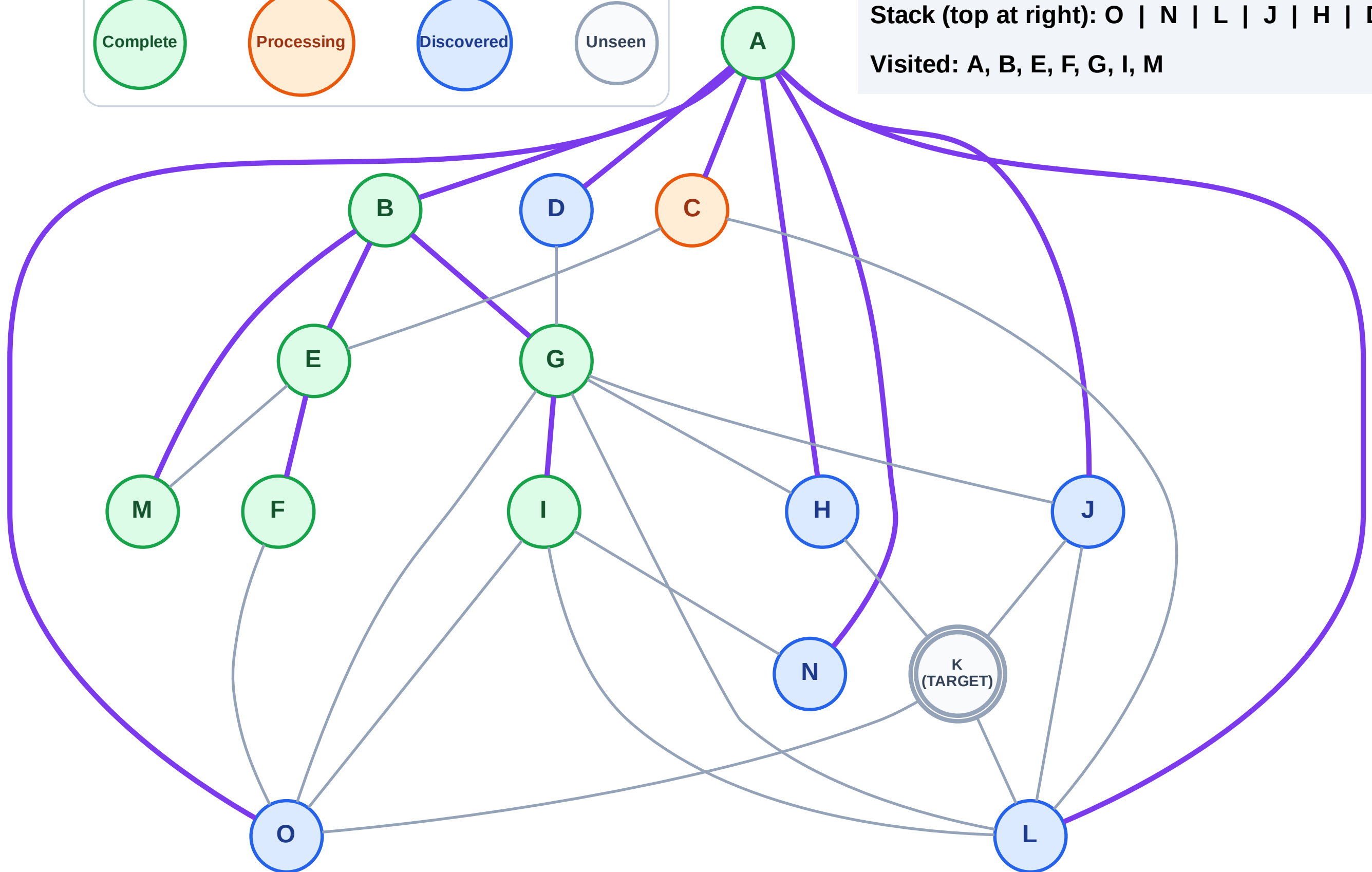
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D

Visited: A, B, E, F, G, I, M



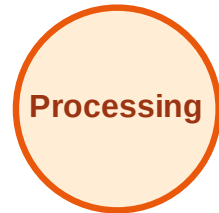
DFS Traversal

Step 17: No new neighbors from C

Legend



Complete



Processing



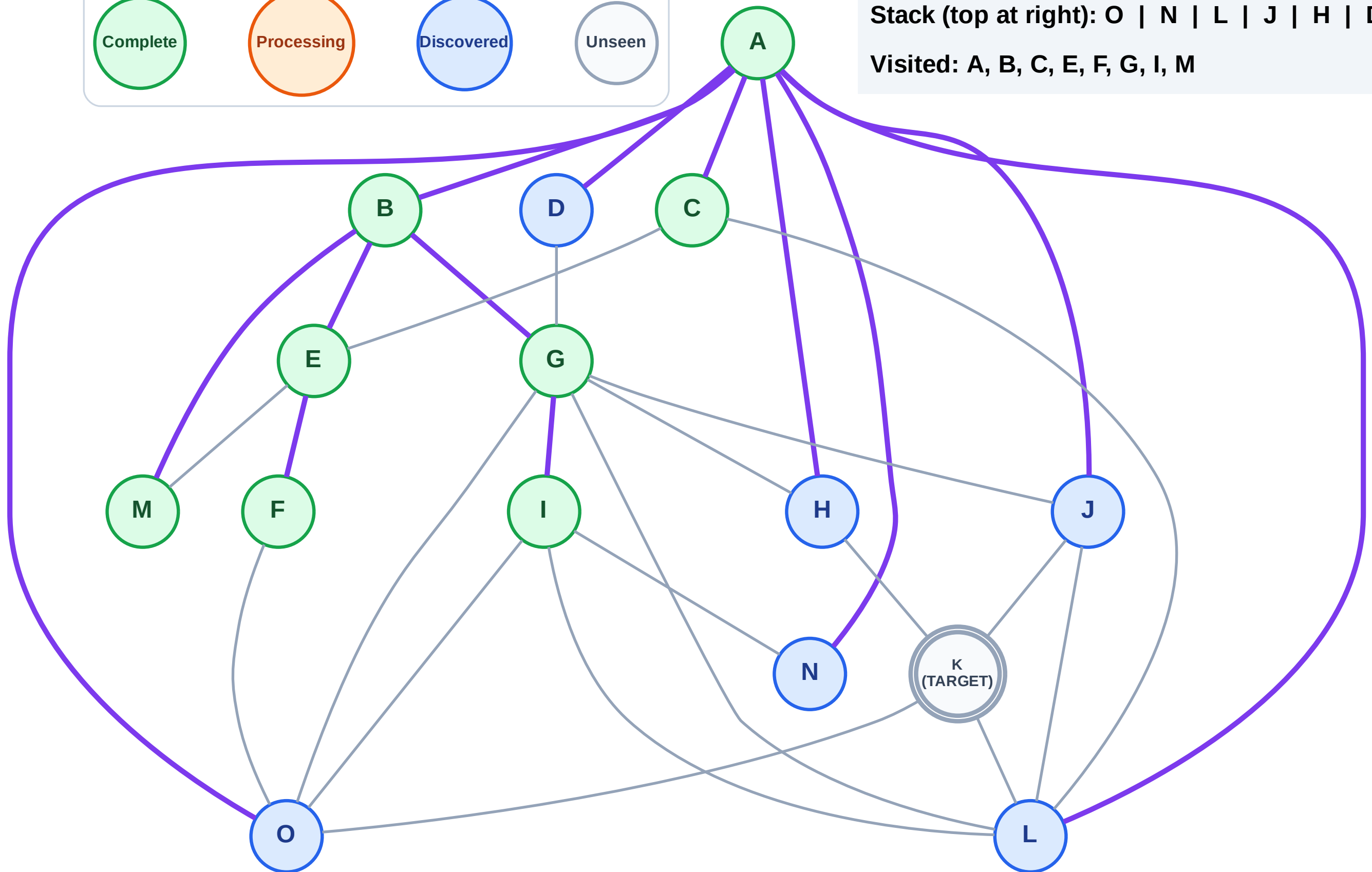
Discovered



Unseen

Stack (top at right): O | N | L | J | H | D

Visited: A, B, C, E, F, G, I, M



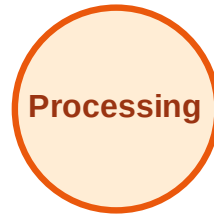
DFS Traversal

Step 18: Process node D

Legend



Complete



Processing



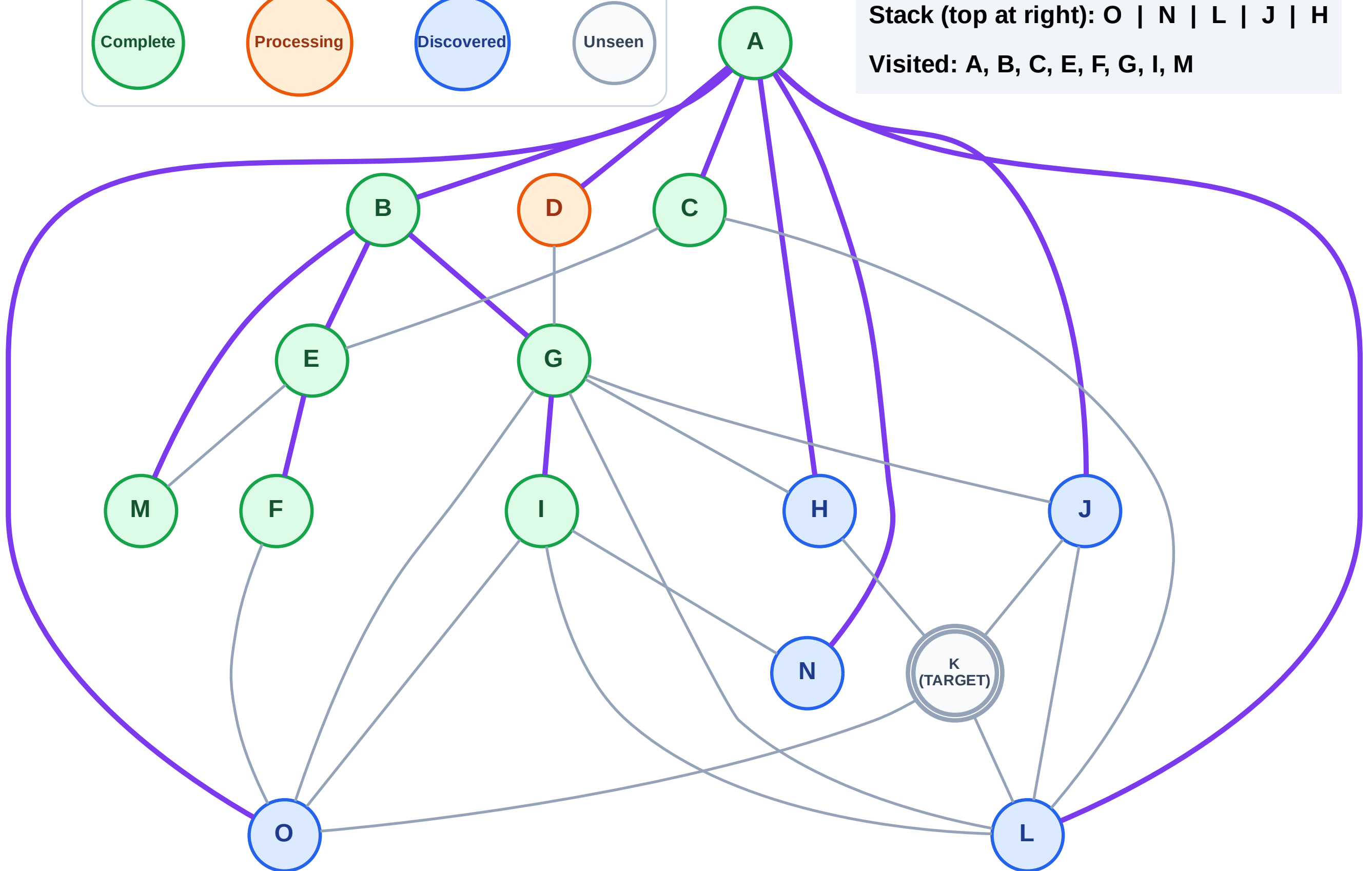
Discovered



Unseen

Stack (top at right): O | N | L | J | H

Visited: A, B, C, E, F, G, I, M



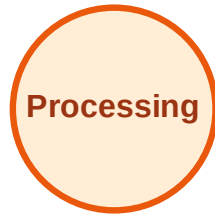
DFS Traversal

Step 19: No new neighbors from D

Legend



Complete



Processing



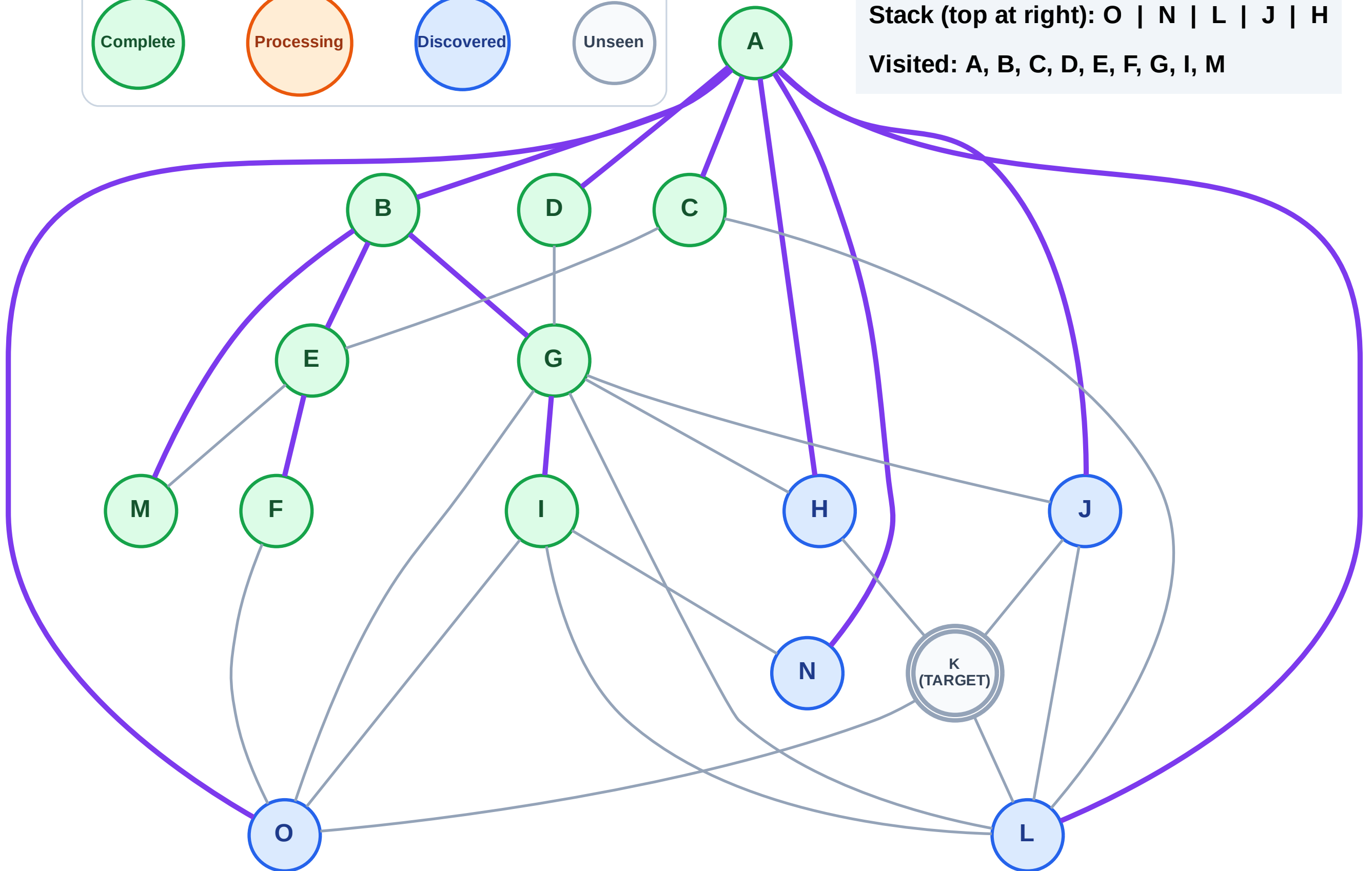
Discovered



Unseen

Stack (top at right): O | N | L | J | H

Visited: A, B, C, D, E, F, G, I, M



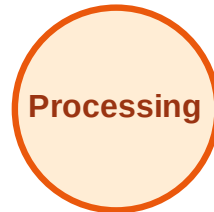
DFS Traversal

Step 20: Process node H

Legend



Complete



Processing



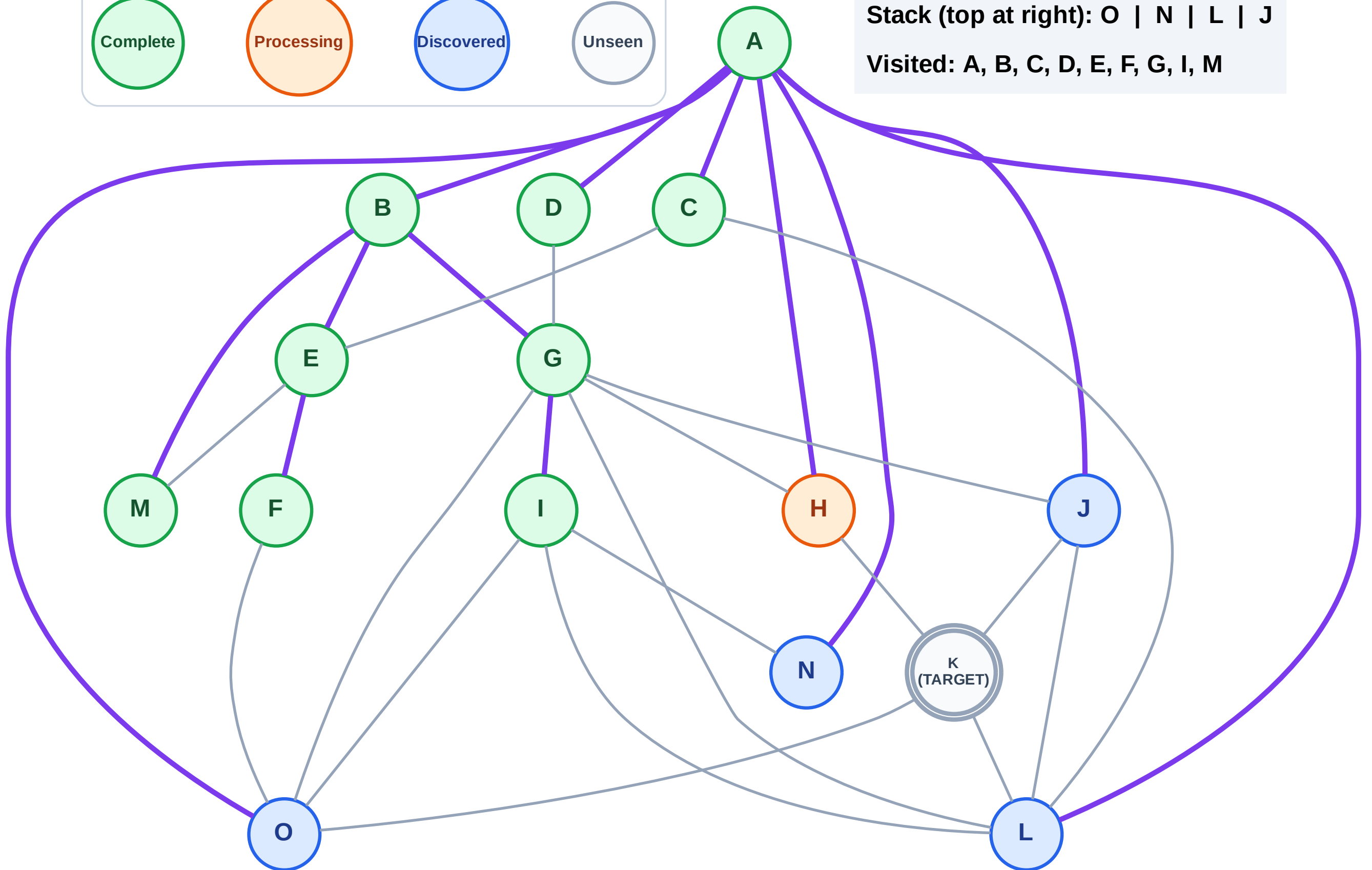
Discovered



Unseen

Stack (top at right): O | N | L | J

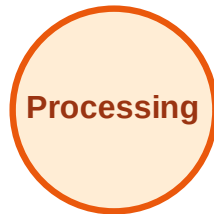
Visited: A, B, C, D, E, F, G, I, M



DFS Traversal

Step 21: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → H → K

Legend



Stack (top at right): O | N | L | J
Visited: A, B, C, D, E, F, G, H, I, K, M

